

Data Operations Guide

For NetBox™ and NetBox Global

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Contents

About This Guide	1
Data Operations Concepts and Overview.....	1
Import Operations	4
Export Operations	5
Global Data Operations.....	6
Best Practices When Importing Person Records	9
Import File Format.....	11
Keyword Header Fields	11
Specifying Primary Person Data	12
Specifying Secondary Person Data	13
Specifying Credential Data.....	15
Specifying Access Level Data	17
Specifying Mobile Apps Data	19
Specifying Badging Data	21
Specifying System Login Data	21
Repeating and Complex Fields	22
Examples	24
Delimiters	30
Date and Time Fields	32
Using the Exclamation Point (!) for Comments	33
Special Character Handling	33
User-Defined Fields	33
Partition Fields	35
Access Levels Duplicated Across Partitions	35
Export Files.....	36
Global and NetBox Export Windows	37
Comparing Data Operations and NetBox Data Management Tool Formats	41
Data Management Tool Fields with Multiple Instances	41
Data Operations Complex and Repeating Fields	42
Configuring Data Operations	43
Uploading an Import File	43
Downloading Results of Past Import Operations	44
Configuring a Network Storage Location for Import Operations.....	44
Scheduling an Automatic Import Operation	45
Downloading an Export File	48
Viewing the Results of Data Operations	49
Error Messages	51
Viewing Global Import File Errors	51
Error Processing and Log Files	52
Examples of Import Files with Errors	53
Data Operations Error Messages.....	53
Appendix A - Attribute Maximum Field Length (Person/Vehicle).....	58

About This Guide

This guide describes how to configure, operate, and monitor the Data Operations feature available in NetBox™ and NetBox Global systems. The intended audience is security administrators who are responsible for managing the import and export of person record data.

The Data Operations feature allows you to use comma-separated and tab-separated Import and Export files to add, delete and modify person records in your NetBox and Global systems. You can:

- Populate your system database with person records.
- Update your person records.
- Produce an Export file that can be accessed by external applications.
- Schedule automatic processing of Import files placed in an accessible NAS location.
- Track all changes made from data import operations.

Important: If Data Operations is used to update person attributes and access level assignments that are managed on an LDAP server, the updates will be overwritten the next time this data is synchronized—either when the data is changed on the LDAP server or when the next full synchronization is performed. For this reason, we do not recommend using Data Operations to update data on a NetBox that synchronizes its data with an LDAP server.

A full synchronization is performed whenever communication with the LDAP server is stopped and restarted and whenever the LDAP server configuration is disabled and re-enabled.

Data Operations Concepts and Overview

The Data Operations feature lets you add person records (including access level, credential, and user-defined person record information) to your NetBox or Global system. You can also modify, delete, and purge existing person records.

With the Data Operations feature, you can:

- Manually upload comma-separated values (CSV) or tab-separated values (TSV) text files. (Import files).
- Manually download CSV or TXT text files (Export files).
- Automatically process an Import file at scheduled intervals from a pre-configured NAS location.

Figure 1 depicts a user or external application creating an Import file from a database or work order external to an NetBox system. The NetBox administrator uses Data Operations to import the file and update person records.

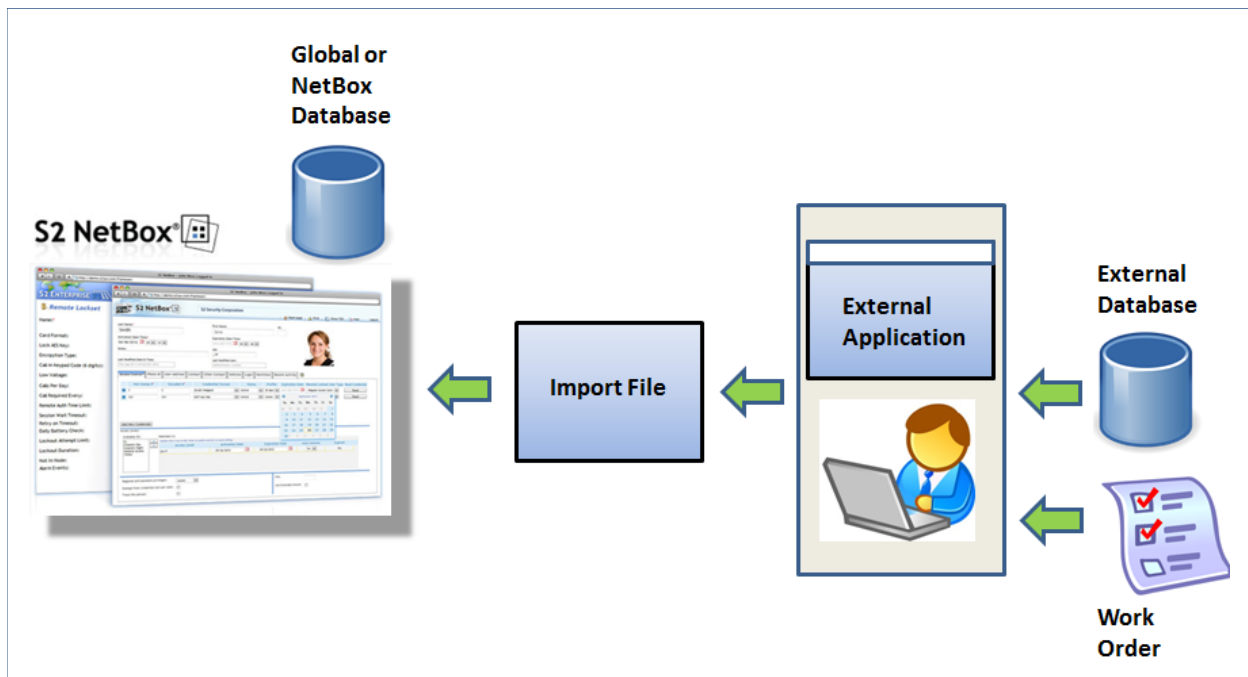


Figure 1: Manual upload of an Import file.

Figure 2 depicts the Data Operations Export feature where an Export file is generated with specific person record data for each employee. A user or external application accesses the Export file and updates a database external to the controller.

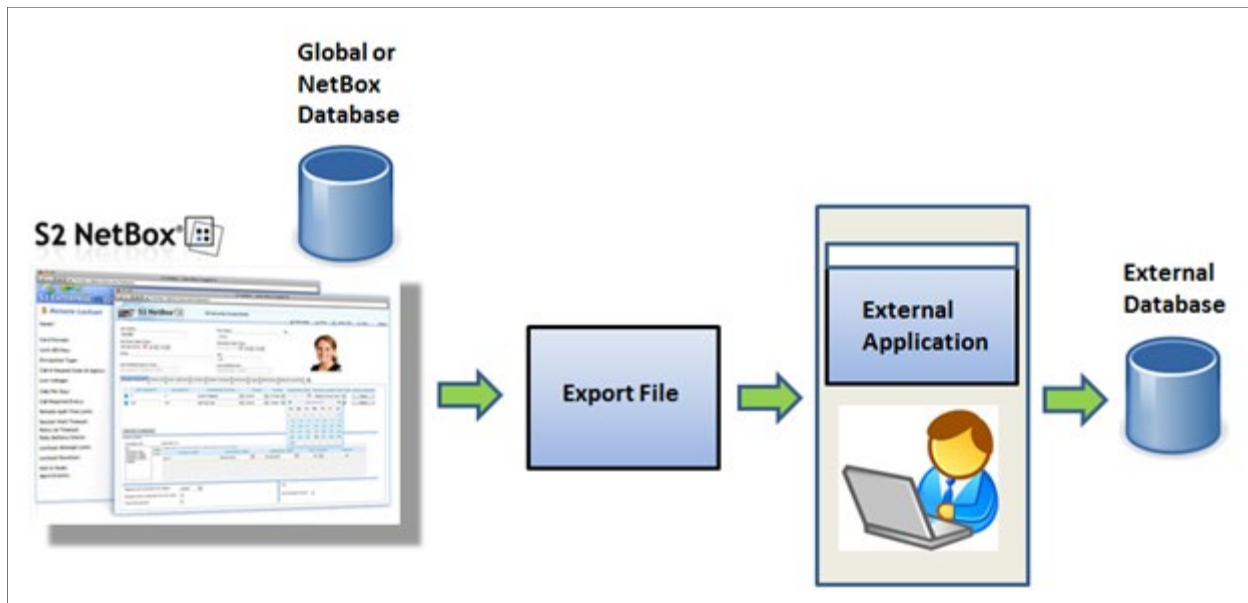


Figure 2: Manual download of an Export file.

Figure 3 depicts an automatic import of a file created by an external application or user and placed in a pre-configured NAS location. Data Operations polls the NAS location at a user-defined interval and updates the person records of a NetBox system.

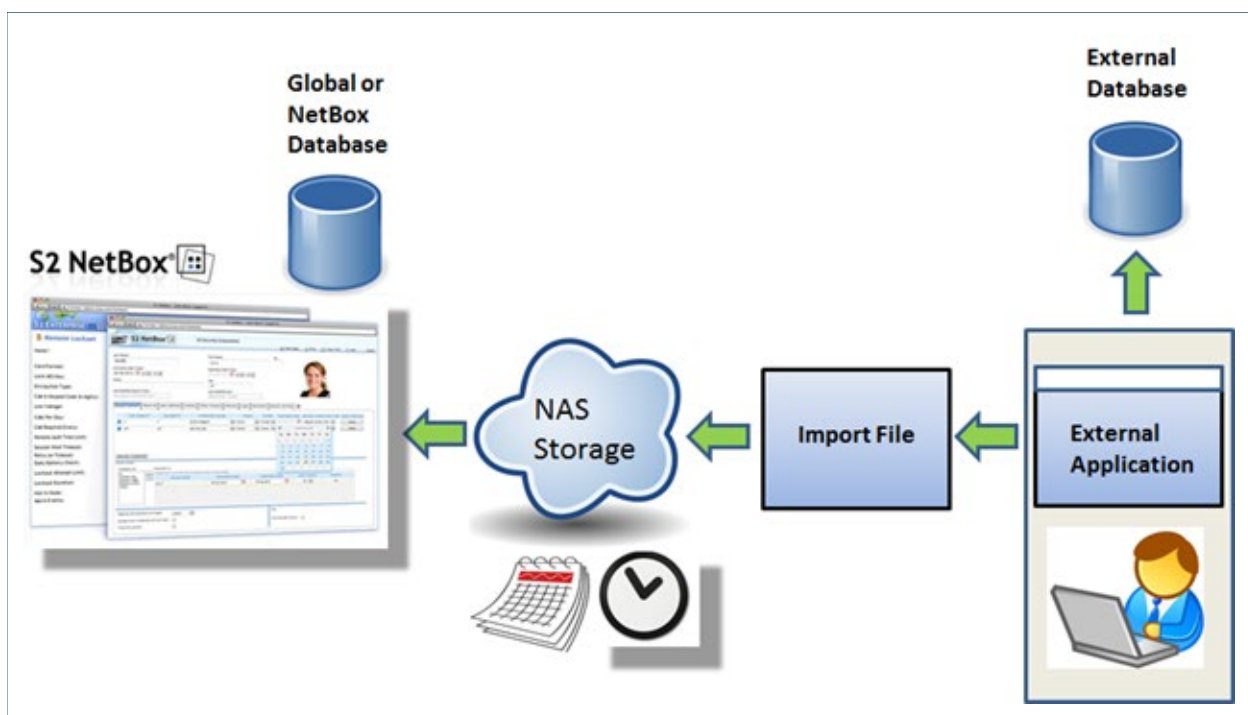


Figure 3: Automatic upload of an Import file.

Figure 4 depicts a sample workflow where an external application is used to import employee data from an external database data on a scheduled basis. The external application processes the import data and updates the NetBox person records at a user-specified interval. The external application also collects export data resulting from changes in the Global and NetBox databases. The export data is used to update the external database.

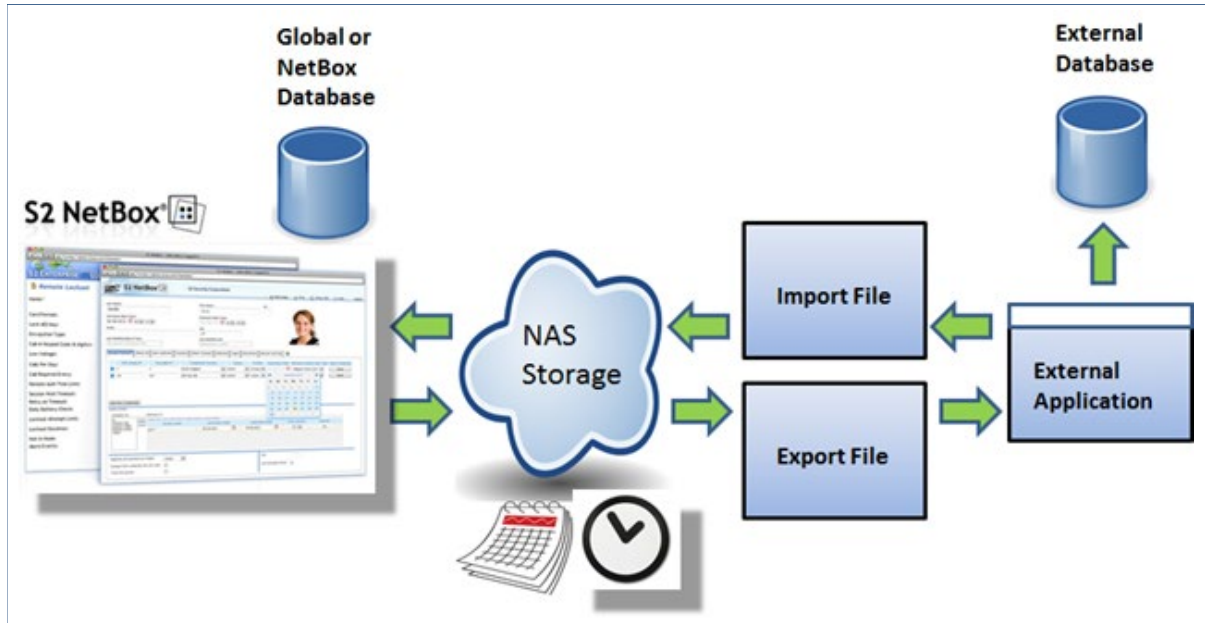


Figure 4: Using an external application to import person record data.

Import Operations

On import, Data Operations reads records from an Import file and updates person records in your NetBox or Global system database. You can use the Data Operations import feature to:

- Populate your system database with person records.
- Modify existing person records.
- Delete existing person records.
- Purge existing person records.

For Global users, Data Operations provides the only means for purging person records. There is no purge person capability in the Global user interface.

- Schedule automatic processing of Import files placed in an accessible NAS location when periodic updates to your person records are required.
- Monitor data import operations for errors using the operating log that is produced for each operation.

Initially Populating Your System with Person Records

When a new system is installed, it is necessary to create persons and card credential data for all persons who will require access. You could enter the data one person record at a time using the Global and NetBox user interface.

An alternative method would be to produce a master Import file that contains all the person record data that will be required to populate your system using an external application such as Microsoft Excel or Access. These programs are designed to expedite and facilitate the data creation and management process. (See section: [Import File Format](#) on page 11 for details on Import file format.)

Once this file has been created, you can use Data Operations to Import the file and populate your system with person records for each employee.

Updating Your Person Records

Periodically, it will be necessary to update your person records with additional data as new hires come on board, employees leave, and modifications are required to existing records. If these updates are infrequent, they can be managed easily via the Global and NetBox user interfaces.

You may be required to do bulk updates on a periodic basis. Data Operations provides an efficient method by which this can be performed:

1. An external application or an individual creates an Import file which contains modifications that will update the person records on your system.
2. You use the Data Operations Import feature to Import the file manually via the Global or NetBox user interface to update person records.

Scheduling Automatic Processing of Import Files

Data Operations allows you to schedule the periodic processing of an Import file from an accessible NAS location on your system at some interval. This is useful when security-related updates need to be generated from an external source (for example, an external application that updates the corporate database with employee information).

The external application generates an Import file and places it in the NAS location. Data Operations polls the NAS location, performs an Import operation, processes the Import file, and updates your Global or NetBox person records with the new data.

Export Operations

On export, Data Operations downloads an Export file that contains all the person record data in your Global or NetBox system to a location that you select via your browser where it can be accessed by external software applications.

You can use Data Operations to create an Export file that can be accessed by external data management tools (such as Microsoft Excel or Access) so

that modifications made to your Global or NetBox person records can be made to an external database system.

Global Data Operations

In a NetBox security network that is composed of controllers and nodes and does not have a Global system installed, data operations that update and modify person records on an controller are performed independently of other controllers on the network.

A controller is not aware of data operations processing performed on other controllers in the security network. If central management of person record data is required, it must be managed using external applications that monitor and track changes to independent databases running on each controller.

The addition of a Global system to your system environment makes it possible to manage all your data operations from one centralized location. Global allows you to tie together all your controllers under one centrally-controlled, distributed network and manage them via a web browser.

The Global system, as one of its core features, replicates Credential, Access Level, and person record data to every NetBox system to which it is connected. It also receives updated person record data from each connected controller and adds it to the Global system database.

For example, Acme Chemical's security network is comprised of three nodes: NetBox1, NetBox2 and NetBox3. The modifications are entered in one of the following ways:

- Security or administrative staff manually updates person records using the NetBox or Global user interface.
- Data Operations processes the Import files.
- A user-developed application running externally to the security network performs automated processing.
- Import files are accessed on a scheduled basis at a NAS location.

Figure 5 shows how an external application developed internally by Acme Chemical, called Central Data Manager, is used to communicate with the Global node in the security network to obtain changes that have been made to person record data. The Global system updates each NetBox node with these changes as required—either globally on all nodes in the security network, or on individual nodes.

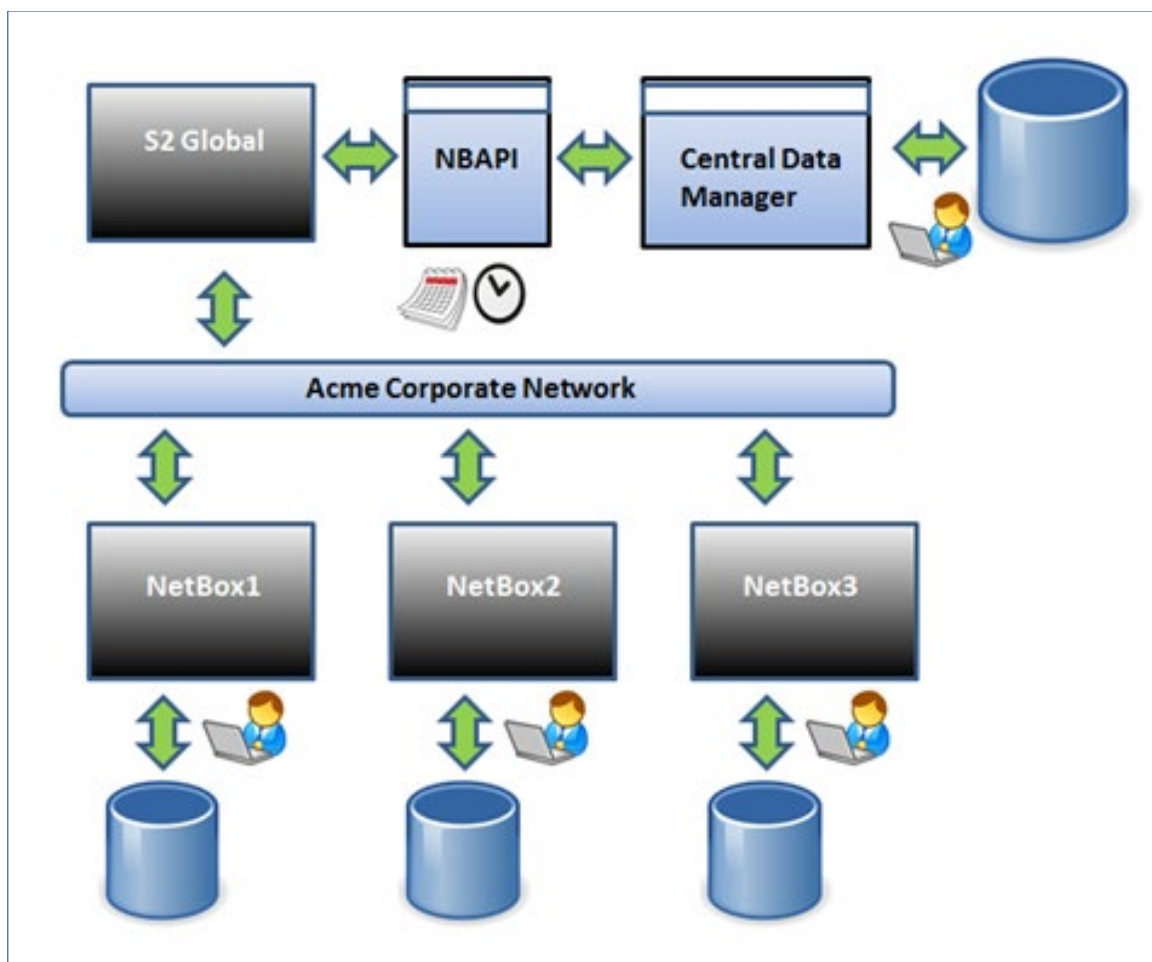


Figure 5: Updating person records in an Global system.

Each day person records across the corporate enterprise change as a result of normal day-to-day operations. New employees come on board, and existing employees leave, change their status, change their vehicles, and so on. These person record changes can occur at various points in an Global system:

- On the Global server.
- On one or more of the controllers.
- Outside the LenelS2 security network from an internal department such as HR or Security.

For example, NetBox1 has 2 new employees starting at the beginning of the week and one going on extended leave (as shown in Figure 6). Several employees have new cars to register and a new access level must be added because a new laboratory has been opened with 10 employees requiring updates to the credential and access levels in their person records. The person records are updated manually by the security administrator during the day.

NetBox2 and NetBox3 also have person record updates that are entered during the day by security administrators.

There are also changes to the corporate employee database outside the LenelS2 security network. Personal employee data records in the corporate database have been updated by the Human Resource and Security departments which also will require updates of person records on nodes in the LenelS2 security network.

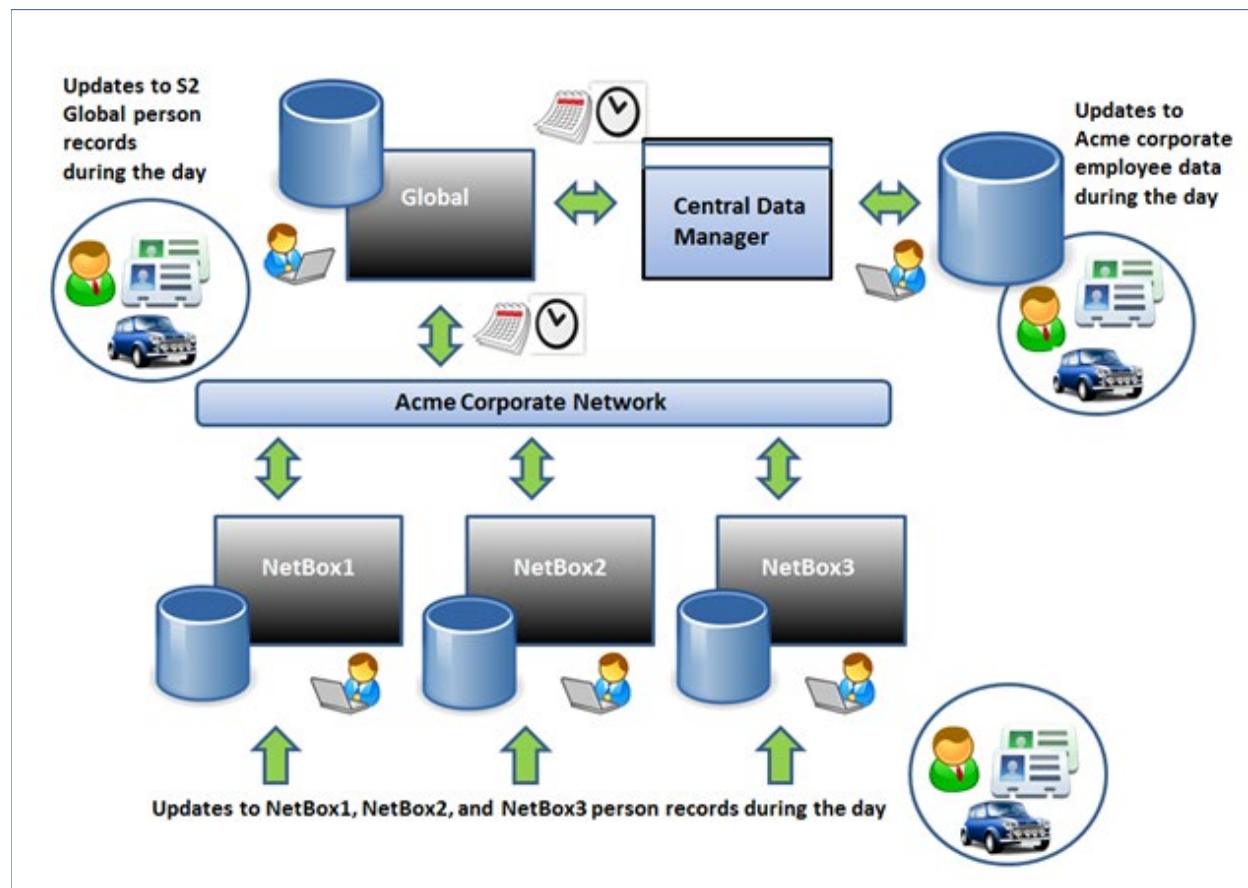


Figure 6: Daily corporate employee data updates by Acme Chemical.

Throughout the day, Global system updates the person records in its database with changes received from the three controllers. At the end of the day, an Export file is created by a security administrator and placed in a specific NAS location known to the Central Data Manager. This Export file contains updates for all data fields for every person record in the Global database.

The Central Data Manager reads the Export file data from the Global node and updates the corporate central database. It also places an Import file of its own in a specified NAS location so that the Global system can access the file, process its contents and update its person record data. When the Global database is updated, the Global system updates each NetBox node.

Figure 7 shows the data flow between the Global system and the Central Data Manager at the end of the day, which updates the security system person records and the employee records in the Acme Chemical corporate database.

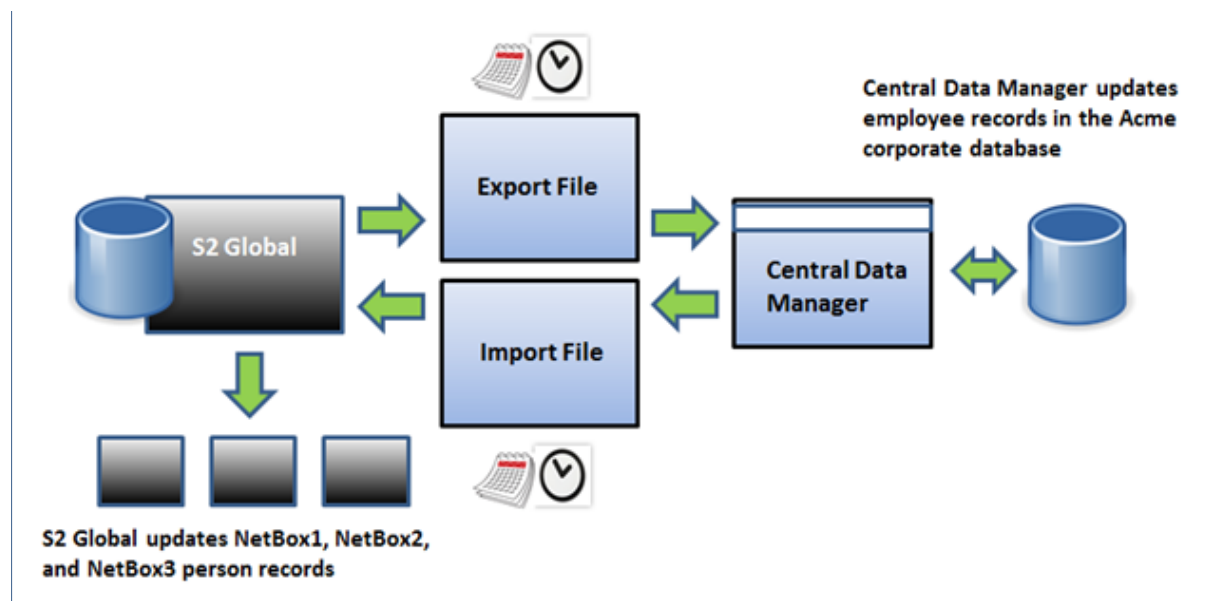


Figure 7: Final update of security system person records and the Acme Chemical corporate database.

The example above describes one simple scenario. There are an many other possibilities for the exchange of person record data and data external to the LenelS2 security network, which are made possible by having an Global system running on your security network.

Best Practices When Importing Person Records

Specifying Partitions in NetBox Import Files

By default, each NetBox has a Master partition, and one or more additional partitions can be added to the system.

When adding person records via an Import file to a NetBox with multiple partitions, you will need to specify the partition to which the person will belong. See: [Import File Format](#) on page 11 for details on specifying the partition when creating an Import file.

Specifying Controllers and Partitions in Global Import Files

When a new partition is added to a NetBox connected to an Global system, Global recognizes the partition automatically and adds a new Controller tile to the Partitions page (*Monitor : Partitions*).

Note: In Global, NetBox systems are referred to as Controllers.

For example, Figure 8 shows the Partitions page for an Global system to which a NetBox named Tech Pubs Quatro has been added.

In addition to an Global partition tile, the page displays Controller partition tiles for the NetBox Master partition and two user-defined partitions—New Corp and Third Corp Financial:

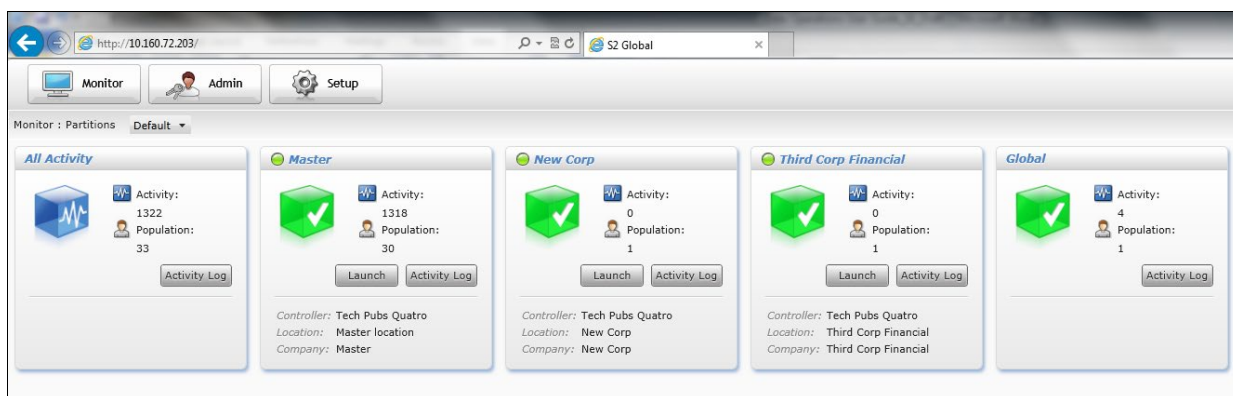


Figure 8: Partition tiles in Global.

When adding person records via an Import file to an Global system with multiple NetBox systems (Controllers) configured, it is strongly recommended that you specify the both Controller and the partition in every import file.

This will allow Data Operations to differentiate partitions should multiple Controllers have partitions with the same name, as in this example:

The screenshot shows the NetBox Global interface with a table of partitions. The table has columns for Type, Name, Controller, Location, Company, and Description. A red box highlights the 'New Corp' partition under the 'Controller' column.

Type	Name	Controller	Location	Company	Description
Global	Global		Global	Global	Unmanaged Location
Controller	Master	Tech Pubs	Master location	Master	The default partition
Controller	Master	Tech Pubs 2	Master location	Master	The default partition
Controller	New Corp	Tech Pubs	New Corp	New Corp	Merger with New Corp
Controller	New Corp	Tech Pubs 2	New Corp	New Corp	
Controller	Third Corp Financial	Tech Pubs	Third Corp Financial	Third Corp Financial	Acquisition of Third Corp Financial 9_01_2013

Figure 9: Duplicate partitions configured on different controllers.

See: [Import File Format](#) on page 11 for details on specifying the Controller and partition when creating an Import file.

Import File Format

A Data Operations Import file is a comma-separated values (CSV) or tab-separated values (TSV) text file. The file begins with a row of header fields, followed by one or more rows of data fields.

Each header field contains a keyword that indicates the type of data specified in that column. For example, the first header field always contains the keyword **Command**. For a data row to be processed, the value in the Command column must be one of the following import commands:

AddPerson, **ModifyPerson**, or **ModifyOrAddPerson**.

Keyword Header Fields

Note: Person and vehicle attribute maximum field length, see “Appendix A - Attribute Maximum Field Length (Person/Vehicle)” on page 58.

This section lists the keyword header fields that can be used in an Import file to specify the operation to be performed, where updates should be applied, and the following person record data:

- [Primary person data](#) – Appears at the top of a person record in the NetBox or Global web interface.
- [Secondary person data](#) – Appears on the Information tab.
- [Credential data](#) – Appears on the Credentials tab.
- [Access level data](#) – Appears on the Access Levels tab.
- [Mobile apps data](#) – Appears on the Mobile Apps tab.
- [Badging data](#) – Appears on the Badging tab.
- [System login data](#) – Appears on the Login tab.

Notes: Keywords are not case-sensitive, but they are normalized to uppercase during processing.

In a data row, values are not required for all columns. If a column is specified but a data row includes no value for that column, that field is left unmodified in the person record.

To clear a field value from a person record, enter the value **<%BLANK%>**.

The keyword header fields and header subfields are text fields unless otherwise specified.

Specifying the Import Operation and Where Updates Should Be Applied

To specify the import operation to be performed and where updates should be applied, use the keyword header fields listed in the following table.

Table 1: Keyword Header Fields for Specifying the Operation to Be Performed and Where Updates Should Be Applied.

Keyword Header Field	Usage Notes
Command	Required. This must be the first keyword in the header row. For a data row to be processed, the value in this column must be one of the following commands: • AddPerson: Adds a new person record. • ModifyPerson: Modifies an existing person record. • ModifyOrAddPerson: Adds a new person record or modifies an existing person record. If the system cannot find an existing person record with the specified PersonID, it adds a new person record with that Person ID.
Controller	The name of the NetBox where updates will be applied. If you are importing data into an Global system, use the name \$GLOBAL to apply the updates to the Global system.
Partition	The name of the NetBox partition where person record updates will be applied. If you are importing data into an Global system use the name \$GLOBAL to apply the updates to the Global partition.

Specifying Primary Person Data

To specify primary person data to be added or modified, use the keyword header fields listed in the following table. The updates will appear in fields at the top of the new and modified person records.

Table 2: Keyword Header Fields for Specifying Primary Person Data.

Keyword Header Field	Usage Notes
ActivationDate	The date or date/time a person record will become active. If this information is not supplied in an AddPerson data row, the current date and time are used. A date or date/time is specified in the following format: • YYYY-MM-DD • YYYY-MM-DD Valid dates are in the range: 1970 or later.

Keyword Header Field	Usage Notes
Deleted	Requires a Boolean value that is either TRUE or FALSE. TRUE deletes the person record, which removes it from the active roster but not from the system.
ExpirationDate	The date or date/time when a person record will expire. A date or date/time is specified in the following format: • YYYY-MM-DD • YYYY-MM-DD Valid dates are in the range: 1970 or later.
FirstName	The person's first name.
LastName	The person's last name. This is required for the AddPerson , command. It is also required for the ModifyorAddPerson command if the person record does not already exist.
MiddleName	The person's middle initial.
NewPersonID	Used with the ModifyPerson command to change the person's ID number.
Notes	Additional information to appear in a text field at the top of the person record.
PersonID	The person's ID number. This is required for all commands because it is used to uniquely identify a person within the system.
PictureFile	The name of a picture file stored for input in a NAS storage location for automatic import operations. Note that Data Operations can upload pictures only through automatic import operations. See Scheduling an Automatic Import Operation on page 45 for information on setting up directories and sub-directories for Data Operations.
Purge	Requires a Boolean value that is either TRUE or FALSE. TRUE purges the person record, which permanently removes the record and all related data from the system.

Specifying Secondary Person Data

To specify secondary person data to be added or modified, use the keyword header fields listed in the following table. The updates will appear on the Information tab in the new and modified person records.

Note: Person and vehicle attribute maximum field length, see “Appendix A - Attribute Maximum Field Length (Person/Vehicle)” on page 58.

Table 3: Keyword Header Fields for Specifying Secondary Person Data

Keyword Header Field	Usage Notes
UDF1 through UDF20	Names corresponding to user defined fields (UDFs) defined in your system. For each UDF, the specified values must match the defined field type: Text, Numeric, Boolean, or List.
Contact section:	
Email	The person's primary email address.
Email2	The person's SMS address.
Phone	The person's primary phone number.
Location	The person's location.
MobilePhone	The person's mobile phone number.
Other Contact section:	
OtherContactName	A secondary contact person.
OtherContactPhone1	A secondary phone number for a contact person.
OtherContactPhone2	A tertiary phone number for a contact person.
Vehicles section:	
Vehicles [/isf]	A list of vehicles on record for the person. Vehicles is a complex field.
Vehicles Subfields:	
Color	The vehicle's color.
Licnum	The vehicle's state-issued license plate number.
Make	The vehicle's manufacturer.
Model	The vehicle's model name and/or number.
State	The standard two-character code for the state where the vehicle is registered.
Tagnum	The company-issued parking permit number.

Specifying Credential Data

To specify credential data to be added or modified, use the keyword header fields listed in the following table. The updates will appear on the Credentials tab in the new and modified person records.

Table 4: Keyword Header Fields for Specifying Credential Data.

Keyword Header Field	Usage Notes
Credentials [/isf]	A list of credentials issued to the person. A person record cannot have two different credentials with the same credential format and the same bits. Credentials is a complex field.
Credentials Subfields:	
CardFormat	A credential format based on the credential format names defined in the system.
EncodedNum	The encoded number assigned to a credential.
ExpirationDate	The date or date/time on which the credential's status will change from Active to Expired. The date or date/time is specified in the following format: • YYYY-MM-DD • YYYY-MM-DD If <i>Enable credential expiration requirement</i> is selected for the controller, an expiration date is required for all new and modified credentials.
HotStampNum	The Hot Stamp number assigned to a credential.
Profile	(optional) The name of the credential profile assigned to the credential. The profile defines the maximum number of days of non-use before the credential will expire

Keyword Header Field	Usage Notes
RemoteLocksetUserType	Applicable only for systems with ASSA ABLOY remote locksets or Allegion Engage Wireless Locksets. Used to specify the user type for a credential. The default is blank, which is the same as Regular Access Card. Valid values are: • Regular Access Card • Trigger Comm Update • Toggle Passage Mode • Relock Passage • Panic Lockdown • Lockout Users • Master Clear Lockouts • Emergency Open • Supervisor Open Note: Only Regular Access Card, Emergency Open, or Toggle Passage Mode can be specified for an Allegion Engage Lockset.
Status	The status of a credential, which can be one of the following: • Active • Lost • Suspended • Clear • Not Returned • Missing (Active) • Damaged • Not Used • Missing (Disabled) • Disabled • Not Validated • Temporary • Expired • Returned • Temporary (Expired) • Forgotten • Stolen Note: The last four states may appear on export, but they cannot be set using your system user interface or the NetBox API.

Keyword Header Field	Usage Notes
Settings section:	
AntiPassBackPriv	Used to specify the person's anti-passback privileges. These determine how the system will handle the person's passback and tailgate violations. Valid values for NetBox are: Exempt, Soft Always, and Hard Always. Valid values for Global are: Exempt, Soft, and Hard.
ExemptFromNonUse	Used to exempt the person from credential non-use rules. Requires a Boolean value that is either TRUE or FALSE. If TRUE, the person is exempted from the system's credential non-use rules. If the field is left blank, the value is changed to FALSE.
TracePerson	Used to specify the person's activity should be traced. Requires a Boolean value that is either TRUE or FALSE. If TRUE, a message will appear in the Activity Log whenever the person makes a valid or invalid access request in the active partition. If the field is left blank, the value is changed to FALSE.
UseExtendedUnlock	Used to assign an extended unlock period that will allow the person additional time to get through a door. Requires a Boolean value that is either TRUE or FALSE. If TRUE, any portal accessed by the person will remain unlocked for the number of seconds specified by the Extended Time setting in the portal definition. If the field is left blank, the value is changed to FALSE.
PIN section:	
PIN	Used to assign a PIN, or personal identification number, to the person. Requires a four- to six-digit number.
ExemptFromPin	Used to exempt the person from the need to enter a PIN at portals with keypads. Requires a Boolean value that is either TRUE or FALSE. If TRUE, the person will have card-only access at portals with keypads.

Specifying Access Level Data

To specify access level data to be added or modified, use the keyword header fields listed in the following table. The updates will appear on the Access Levels tab in the new and modified person records.

Table 5: Keyword Header Fields for Specifying Access Level Data.

Keyword Header Field	Usage Notes
AccessLevels <i>[list]</i>	A list of access levels and access level groups assigned to the person. AccessLevels is a complex field.

Keyword Header Field	Usage Notes
Header Subfields:	
AccessLevelName	The name defined in the system for the access level or group.
ActivationDate	The date or date/time when the access level or group will become active. The date or date/time is specified in the following format: • YYYY-MM-DD • YYYY-MM-DD If no activation date is specified or you enter <%BLANK%> in the field, an activation date will not be defined in the system.
AutoRemove	Indication of whether the access level will be removed from the person record automatically once it expires. Requires a Boolean value that is either TRUE or FALSE. If TRUE, the access level will be removed once it expires. An access level that is removed automatically will still exist in the system, but it will no longer be assigned to the person.
ExpirationDate	The date when the access level will expire. The date is specified in the following format: • YYYY-MM-DD • YYYY-MM-DD If no activation date is specified or you enter <%BLANK%> in the field, no expiration date will be defined in the system.

Specifying Mobile Apps Data

To specify the mobile apps data to be added or modified, use the keyword header fields listed in the following table. The updates will appear on the Mobile Apps tab in the new and modified person records. When adding or modifying mobile apps data, note that:

- For a person record to be added or modified successfully, the person's mobile phone number and primary email address must be filled in on the Information tab of the person record. Use the MobilePhone and Email keywords to fill in these fields via an Import operation.
- If Mobile Security User™ is not enabled for the system, the Mobile Apps settings will take effect only after it has been enabled.

Table 6: Keyword Header Fields for Specifying Mobile Apps Data.

Keyword Header Field	Usage Notes
MSUEnabled	Used to enable or disable Mobile Security User for the person. Requires a Boolean value that is either TRUE or FALSE. TRUE enables Mobile Security User.
BlueDiamondEnabled	Used to enable or disable BlueDiamond Mobile Credentials for the person. Requires a Boolean value that is either TRUE or FALSE. TRUE enables BlueDiamond Mobile Credentials.
MobileCredentialRequest	(optional) Used to add (+) a mobile credential to a person or remove (-) a mobile credential from a person. Requires the name of a mobile credential format of a mobile credential bundle already assigned to this NetBox. The person cannot have any other mobile credential pending or active from any mobile credential bundle of the same type to add a mobile credential using Data Operations. Note: Credentials and MobileCredentialRequest columns cannot be used in same row for a person. If a person is being added credentials using the Credentials column, the MobileCredentialRequest column should be populated in a later row for the same person and vice-versa.
PeopleGroups	The names of the people groups to which the person is assigned. Each people group defines a notification process for mobile apps users.
ThreatLevelEnabled	Used to enable or enable the Threat Level Module for the person. Requires a Boolean value that is TRUE or FALSE. TRUE enables the Threat Level Module.
ThreatLevelLocation	Used to specify the location for which a person can escalate the threat level. The location must be defined on the NetBox.
ThreatLevelPhone	(optional) Used to specify the phone number to be used for emergency calls placed from a person's mobile phone.
ThreatLevelPhoneOpt	(optional) Used to specify the level of assistance Mobile Security User will provide for placing emergency calls. The options are None, Autodial, or Manual.

Keyword Header Field	Usage Notes
ThreatLevelSeverity	Used to specify the threat level to which a person can set the threat level location. The threat level must be defined on the NetBox.
VirtualIDLayout	Used to specify the layout name for the badge template layout to be used for a person's virtual photo ID badge. The layout must be defined on the NetBox.

Examples

The following example shows how to add a mobile credential to a person:

```
COMMAND,PERSONID,PARTITION,LASTNAME,FIRSTNAME,MOBILECREDENTIALREQUEST
ModifyPerson,45,Master,L,Ray,{LenelS2 48bit Corp 1000}
```

	A	B	C	D	E	F	G
1	COMMAND	PERSONID	PARTITION	LASTNAME	FIRSTNAME	MOBILECREDENTIALREQUEST	
2	ModifyPerson	45	Master	L	Ray	{LenelS2 48bit Corp 1000}	
3							
4							
5							

Figure 10: Adding a mobile credential to a person.

The following example shows how to remove a mobile credential from a person:

```
COMMAND,PERSONID,PARTITION,LASTNAME,FIRSTNAME,MOBILECREDENTIALREQUEST
ModifyPerson,45,Master,L,Ray,-{LenelS2 48bit Corp 1000}
```

	A	B	C	D	E	F
	COMMAND	PERSONID	PARTITION	LASTNAME	FIRSTNAME	MOBILECREDENTIALREQUEST
	ModifyPerson	45	Master	L	Ray	-{LenelS2 48bit Corp 1000}

Figure 11: Removing a mobile credential from a person.

Specifying Badging Data

To specify the badging data to be added or modified, use the keyword header fields listed in the following table. The updates will appear on the Badging tab in the new and modified person records.

Table 7: Keyword Header Fields for Specifying Badging Data.

Keyword Header Field	Usage Notes
BadgeLayout	Used to specify the file name for the template layout to be used for a person's Photo ID badge. The layout must be defined on the NetBox. Note: The filename must not include the extension. For example, Default01 instead of Default01.idpz.
BadgePrintRequested	Used to specify whether a person's badge should be printed. Requires a Boolean value that is either TRUE or FALSE. TRUE queues the person's badge for printing. This field allows users to add multiple imported person records to the badge workflow, so photo ID badges can be printed without the need to edit each record individually.
PhotoRequested	Used to specify whether a person requires a photo image upload or capture. Requires a Boolean value that is either TRUE or FALSE. TRUE displays an indicator on the Badge Workflow page that a photo image upload or capture is required for the person.

Specifying System Login Data

To specify the system login data to be added or modified, use the keyword header fields listed in the following table. The updates will appear on the Login tab in the new and modified person records.

Table 8: Keyword Header Fields for Specifying System Login Data.

Keyword Header Field	Usage Notes
LoginUserName	The login username for a system user.
LoginUserPassword	The login password for a system user.
LoginUserRole	The user role access assigned when a system user logs into the system. LoginUserName and LoginUserPassword are required fields when assigning a login user role via an import.

Repeating and Complex Fields

A complex field has multiple subfields whereas a simple field contains a single data value per data row (instance). Complex subfields are separated by the tilde (~) character. The order of subfields is predefined and cannot be modified.

Data Operations supports three complex fields:

- Credentials
- AccessLevels
- Vehicles

Repeating fields are fields that have multiple instances in a data row that are delimited by the pipe character (|).

Repeating fields have an optional modifier that specifies how a data field in the Import file will be applied to a target (as shown in Table 9).

Table 9: Repeating Field Modifiers.

Modifier	Result in the Target
+	The source data is added to the target.
-	The same data is removed from the target.
=	The target is replaced by the source data.
No modifier	Same as the equal sign (=).

All values are pre and post-trimmed of spaces. Fields that contain lists require open ({} and closing (}) braces.

Using an Asterisk (*) with the AccessLevels and Credentials Complex Fields

Processing of a data row will fail if both of these conditions are present:

- An Import file contains a data row with the **ModifyPerson** or **ModifyOrAddPerson** command
- An access level or credential that is not defined in the existing person record, but defined in your system

An error message indicates that the access level or credential was not assigned to that person record.

For example, the person record for employee John Smith only has the Lab_Bio1 access level defined. Lab_Bio2 is a valid access level in the system, but it is not assigned to the person record of John Smith.

The Import file includes the following header and data row without an asterisk preceding the **AccessLevels** complex field:

```
COMMAND,PERSONID,PARTITION,ACCESSLEVELS,FIRSTNAME,LASTNAME,PIN
ModifyPerson,2005,Master,{Lab_Bio1~2014-10-23~~FALSE|Lab_Bio2~2014-
10-22~~FALSE},John,Smith2005,1005
```

The data row specifies that the **ExpirationDate** is to be modified for the access level, Lab_Bio2.

When the system encounters the data row and finds that the John Smith person record does not include the Lab_Bio2 access level, processing of that data row fails and an error message indicates the access level was not found for that person record.

To avoid processing errors when a matching access level or credential is not found for a person record, you can place an asterisk (*) in front of the opening brace ({) of the **AccessLevels** and **Credentials** complex fields in the Import file. If the system encounters an access level or credential that is not assigned to the person record, it will add the access level and/or credential to the person record.

Note: The system will add an access level or credential to the person record only if the access level or credential already exists in the system. See the online help for information on creating access levels and credentials in the online help (search for *Setting Up Access Levels*).

For example, the person record for employee John Smith only has the Lab_Bio1 access level defined. Lab_Bio2 is a valid access level in the system, but it is not assigned to the person record of John Smith.

The Import file includes the following header and data row with an asterisk preceding the **AccessLevels** complex field:

```
COMMAND,PERSONID,PARTITION,ACCESSLEVELS,FIRSTNAME,LASTNAME,PIN
ModifyPerson,2005,Master,*{Lab_Bio1~2014-10-23~~FALSE|Lab_Bio2~2014-
10-22~~FALSE},John,Smith2005,1005
```

The data row specifies that the **ExpirationDate** is to be modified for the access level, Lab_Bio2.

When the system encounters the data row and finds that the John Smith person record does not include the Lab_Bio2 access level, it adds the Lab_Bio2 access level to the person record.

If the asterisk is not present, the system will find that the Lab_Bio2 access level is not assigned to the John Smith person record and processing of the data row will fail.

Note: You may not want the system to assign the access level or credential to a person record when it is not present in the Import file data row. You may want the processing of the data row to fail so that you have the option of modifying the Import file row if you determine that it is appropriate. In that case, do not include the asterisk before a complex field.

Using an Asterisk (*) with the Vehicle Complex Field

When you include an asterisk (*) before an **AccessLevels** complex field with the **ModifyPerson** or **ModifyOrAddPerson** command, the system attempts to match the access levels in the complex field in your Import file with access levels defined on your system. If it finds a match it will leave the data unchanged in the person record. If it does not find a match, it will add the access level.

For a **Credentials** complex field, the system searches to find matching encoded numbers and card formats defined on your system.

For a **Vehicles** complex field, because none of the six header subfields in the **Vehicles** complex field have pre-defined keywords, adding the asterisk (*) is meaningless. The result with or without an asterisk (*) is the same. The system will add the data in the Vehicle complex field to the person record.

Examples

The examples in this section show you how to create an Import file to update person records on Global and NetBox systems. They are presented with increasing levels of complexity, starting with a simple **AddPerson** command on a NetBox system and eventually graduating to an Import file that contains header and data rows with all three complex fields and repeating subfields.

Note: If you have an Global system running in your security network, you will most likely have multiple controllers running with multiple partitions. It is good practice and highly recommended that you include a Controller and a Partition column in your Import file to differentiate between controllers with duplicate partitions. This is because you can configure duplicate partition names on different controllers. While this precaution might not always be necessary, a consistent practice of including the Controller and Partition column in all your Import files will reduce the chance of errors, avoid confusion and make troubleshooting easier.

Example: Using CSV Format to Add a Person Record to the Master Partition in a NetBox

This example adds a person record for George Smith with **PersonID** = 2001 to the Master partition of the Tech Pubs 2 NetBox when run through Data Operations.

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME
AddPerson,Tech Pubs 2,Master,2001,Smith,George
```

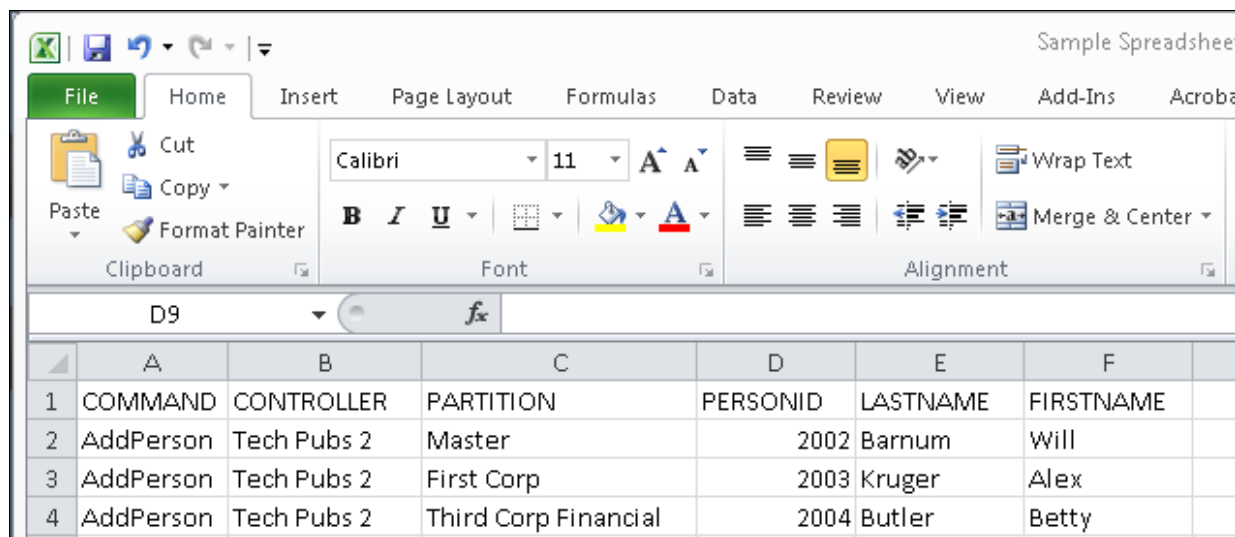
Example: Using TSV Format to Add Person Records to Multiple Partitions in a NetBox

Alternatively, you can use a program such as Excel to create a spreadsheet using the columns in the worksheet for each data field and save it as a TSV (tab-separated variable) file.

This example adds three person records to the NetBox Tech Pubs controller Tech Pubs 2:

- Will Barnum (**PersonID** = 2002) to the Master partition
- Alex Kruger (**PersonID** = 2003) to the New Corp partition
- Betty Butler (**PersonID** = 2004) to the Third Corp Financial partition

Figure shows a sample Excel spreadsheet.



	A	B	C	D	E	F
1	COMMAND	CONTROLLER	PARTITION	PERSONID	LASTNAME	FIRSTNAME
2	AddPerson	Tech Pubs 2	Master	2002	Barnum	Will
3	AddPerson	Tech Pubs 2	First Corp	2003	Kruger	Alex
4	AddPerson	Tech Pubs 2	Third Corp Financial	2004	Butler	Betty

Figure 12: Creating a TSV Import file using an Excel Spreadsheet.

Example: Adding Person Records to an Global System with Multiple NetBox Systems that Have Multiple Partitions

The example below shows the file format of header and data rows in a comma-separated Import file which adds person records to two NetBox systems, Tech Pubs 2 and Bio Labs:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME
AddPerson,Tech Pubs 2,Master,2005,Gilroy,Dab
AddPerson,Tech Pubs 2,New Corp,2006,Dolby,Major
AddPerson,Tech Pubs 2,Third Corp Financial,2007,Butler,Jerry
AddPerson,Bio Labs,Master,2008,Munchin,Ralph
AddPerson,Bio Labs,Pathology Lab,2009,Flinch,Bob
AddPerson,Bio Labs,Bio_Genesis,2010,Dower,Mary
```

The example adds person records to the following Tech Pubs2 partitions:

- Dab Gilroy (**PersonID** = 2005) to the Master partition
- Major Dolby (**PersonID** = 2006) to the New Corp partition
- Jerry Butler (**PersonID** = 2007) to the Third Corp Financial partition

The example adds person records to the following Bio Labs partitions:

- Ralph Munchin (**PersonID** = 2008) to the Master partition
- Bob Flinch (**PersonID** = 2009) to the Pathology Lab partition

- Mary Dower (**PersonID** = 2010) to the Bio_Genesis partition

Example: Adding a Person Record with Credentials Complex Fields

The example below shows the file format of header and data rows that add a person record to a NetBox.

```
COMMAND, PERSONID, LASTNAME, FIRSTNAME, CREDENTIALS
AddPerson, 4001, Master, Karpov, Julia, {3320~3320~26 bit
Wiegand~Active~2014-02-01~}
```

- The header row contains keyword header fields: and are separated by commas:

```
COMMAND, PERSONID, LASTNAME, FIRSTNAME, CREDENTIALS
```

- The data rows contain data fields separated by commas (,):

```
AddPerson, 4001, Master, Karpov, Julia, {3320~3320~26 bit
Wiegand~Active~2014-02-01~}
```

- The complex data fields are enclosed by opening ({) and closing (}) braces:

```
{3320~3320~26 bit Wiegand~Active~2014-02-01~}
```

- The data sub-fields are separated by tildes (~):

```
3320~3320~26 bit Wiegand~Active~2014-02-01~
```

Example: Adding a Person Record with Access Levels Complex Fields

The example below shows the file format of header and data rows in a comma-separated Import file.

```
COMMAND, CONTROLLER, PARTITION, PERSONID, LASTNAME, FIRSTNAME, ACCESSLEVELS
AddPerson, Tech Pubs 2, New Corp, 2012, Hineman, Frank, {Lab_Bio1~~~FALSE|
All_employee~~~FALSE}
```

- The header row contains the keyword header fields and are separated by commas:

```
COMMAND, CONTROLLER, PARTITION, PERSONID, LASTNAME, FIRSTNAME
ACCESSLEVELS
```

- The data rows contains data fields separated by commas (,):

```
AddPerson, Tech Pubs 2, New Corp, 2012, Hineman, Frank,
```

- The complex **AccessLevels** fields are enclosed by opening ({) and closing (}) braces and separated by a pipe (|):

```
{Lab_Bio1~~~FALSE|All_employee~~~FALSE}
```

- The data sub-fields are separated by tildes (~):

```
Lab_Bio1~~~FALSE
```

Example: Adding a Person Record with Vehicles Complex Fields

The example below shows the file format of header and data rows in a comma-separated Import file.

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME,VEHICLES
AddPerson,Tech Pubs 2,New Corp,2012,Hineman,Frank,
{Silver~Bentley~Arnage~MA~RICHDUDE~123|Black~RollsRoyce~Silver
Shadow~NY~MEHAPPY~124}
```

- The header row contains the keyword header fields:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME,VEHICLES
```

- The data rows are separated by commas:

```
AddPerson,Tech Pubs 2,New Corp,2012,Hineman,Frank,
```

- The **Vehicles** complex fields are enclosed by opening ({} and closing (}) braces and separated by a pipe (|):

```
{Silver~Bentley~Arnage~MA~RICHDUDE~123|Black~RollsRoyce~Silver
Shadow~NY~MEHAPPY~124}
```

- The data sub-fields are separated by tildes (~):

```
Silver~Bentley~Arnage~MA~RICHDUDE~123
```

Example: Using a Text file to Modify a Person Record Using Repeating Field Modifiers

The following example modifies the **AccessLevels** data fields by using a plus character (+) to add a new access level to the existing fields in the person record. The **Credentials** data field values are removed from the person record using the minus character (-).

```
COMMAND,PERSONID,ACCESSLEVELS,CREDENTIALS
ModifyPerson,1901,+{Lab_Pathology~2014-02-01~~FALSE},
-{3158~3158~26 bit Wiegand~Active~2014-02-01~~}
```

Example: Using Excel to Modify a Person Record Using Repeating Field Modifiers

If you are entering data into an Excel worksheet you must precede the plus (+) or minus (-) character with a single quote ('). Otherwise, Excel will generate an error, because the plus and minus characters are reserved for entering a formula in a cell.

For example, to add the **AccessLevels** and **Credentials** data fields (as described in the previous example), perform the following steps:

3. Enter the data into the **AccessLevels** data field with a leading single quote.

Single quote (')

	A	B	C	D
1				
2				
3				
4	COMMAND	PERSONID	ACCESSLEVELS	CREDENTIALS
5	ModifyPerson	1901	'+{Lab_Pathology~2014-02-01~FALSE}	
6				

4. Press ENTER. The single quote disappears:

	A	B	C	D
1				
2				
3				
4	COMMAND	PERSONID	ACCESSLEVELS	CREDENTIALS
5	ModifyPerson	1901	+{Lab_Pathology~2014-02-01~FALSE}	
6				

5. Enter the data into the **Credentials** data field with a leading single quote:

Single quote (')

	A	B	C	D
1				
2				
3				
4	COMMAND	PERSONID	ACCESSLEVELS	CREDENTIALS
5	ModifyPerson	1901	+{Lab_Pathology~2014-02-01~FALSE}	'-{3158~3158~26 bit Wiegand~Active~2014-02--01~~}'
6				

6. Press ENTER. The single quote disappears, indicating that Excel has accepted your entry:

	A	B	C	D
1				
2				
3				
4	COMMAND	PERSONID	ACCESSLEVELS	CREDENTIALS
5	ModifyPerson	1901	+{Lab_Pathology~2014-02-01~FALSE}	-{3158~3158~26 bit Wiegand~Active~2014-02--01~~}
6				

If you attempt to enter the data into the data field as described in the previous steps and encounter the Excel error message:

The formula you typed contains an error...If you are not trying to enter a formula, avoid using an equal sign (=) or minus sign (-), or precede it with a single quotation mark (').

Perform the following steps:

1. Click **OK**.
2. Enter a single quote (') before the plus or minus sign in the cell.
3. Press ENTER. The single quote disappears, indicating that Excel has accepted the entry.

Example: Deleting an Existing Person Record

To delete an entire person record from your system, you need to include the **PersonID** and **Delete** header and data fields. The **Delete** data field must be set to "TRUE."

The following example deletes an entire person record with **PersonID** = 1901 from a NetBox (**Controller** = Tech Pubs 2; **Partition** = New Corp):

```
COMMAND,CONTROLLER,PARTITION,PERSONID,DELETED  
ModifyPerson,Tech Pubs 2,New Corp,1901,TRUE
```

Example: Purging an Existing Person Record

To purge a person record and all data related to that record from a NetBox or Global system, you need to include the **PersonID** and **Purge** header and data fields. The Purge header field must be set to "TRUE."

The following example purges a person record with **PersonID** = 2010 from a NetBox (**Controller** = Tech Pubs 2; **Partition** = NewCorp):

```
COMMAND,CONTROLLER,PARTITION,PERSONID,PURGE  
ModifyPerson,Tech Pubs 2,New Corp,2010,TRUE
```

Person records are purged via a batch process, so it will take some time for all data related to a purged person record to be removed from the system.

Important: Purging a person record will not remove related information from archive files—whether they are stored locally on the controller or in an off-controller location, such as an FTP server or network share. Unless you remove this information manually, it could continue to appear in the system; for example, in reports that are configured to search archives.

If a person record is purged from a NetBox that is connected to a Global system, Global will send the purge request to all other connected NetBox systems that are running software version 5.0 or later—and these 5.0 or later systems will also purge the record. If a connected NetBox is running an earlier software version, Global will purge the record from that system when it is upgraded to software version 5.0 or later.

Example: Modifying Multiple Person Records

So far, all the examples in this manual have shown how to create an Import file that adds information for a single person record. The Import file in the following example contains a header row and multiple data rows that add person records. It also contains different **Controller** and **Partition** combinations (**Controller** = Tech Pubs, **Partition** = New Corp and **Controller** = Tech Pubs2, and **Partition** = Third Corp Financial).

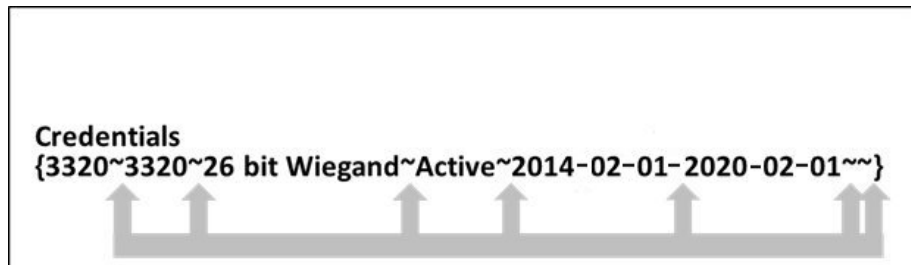
Note: Indented lines are continuations of the same row.

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME,ACCESSLEVELS
AddPerson,Tech Pubs,New Corp,2330,Kandinsky,Wasily,
  Lab_Bio1~~~FALSE|Lab_Bio2~~~FALSE}
AddPerson,Tech Pubs 2,New Corp,2332,Vermeer,Jan,
  {FrontDoorAccess~2013-10-24~2014-10-24~FALSE}
AddPerson,Tech Pubs 2,Third Corp Financial,2333,Klee,Paul
  =(2081~2081~26 bit Wiegand~2013-10-22~Active~Supervisor
  Open|2082~2082~26 bit Wiegand~2014-10-22~Active~Supervisor Open}
```

Delimiters

The following examples show how complex fields: **Credentials**, **AccessLevels** and **Vehicles** are constructed using the tilde (~) and pipe (|) characters as delimiters.

Example: Delimiters for a CREDENTIALS Complex Field with Repeating Fields

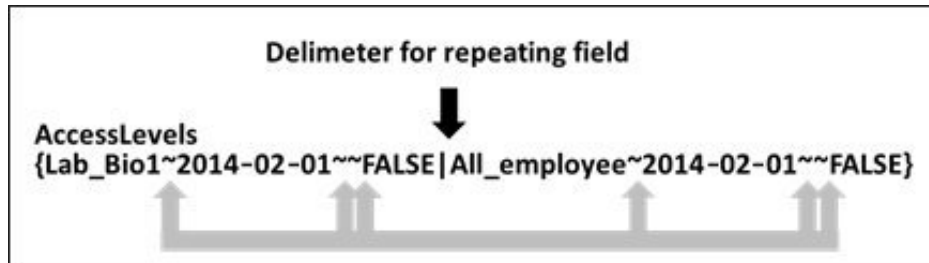


This example of a complex field shows the tildes (~) separating the **Credential** data values. The data values are:

- 3320 (Hotstamp)
- 3320 (EncodedNum)
- 26 bit Wiegand (**CardFormat**)
- Active (**Status**)
- 02-01-2014 (ActivationDate)
- 02-01-2020 (ExpirationDate)

The last delimiter is for the **RemoteLockUserType** subfield, the last subfield in a **Credentials** list. This subfield has no data, so the person record data field **RemoteLockUserType** will not be modified on the system.

Example: Delimiters for an ACCESSLEVELS Complex Field with Repeating Fields



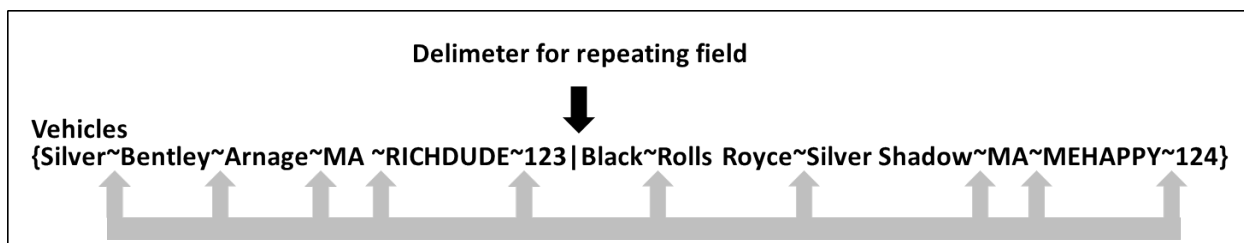
In this example:

- The first two tildes (~) separate the data values for the **AccessLevels** subfields: **AccessLevelName** (Lab_Bio1) and **ActivationDate** (2014-02-01).
- The next tilde (~) (with no data value before it) is a placeholder and specifies that there is no expiration date defined.
 The expiration data will continue undefined until it is changed via the user interface or by an Import data operation.
- The final data subfield specifies the **Auto-Remove** value (TRUE or FALSE) and a tilde (~) delimiter.
 TRUE means that data operations will remove the value and FALSE means the value will not be removed automatically. In this example, the value is FALSE.
- The pipe (|) character specifies that an additional Access Level is defined as part of a list of access levels. In this case there are only two access levels in the list, Lab_Bio1 and All_employee.
 The second access level, All_employee, has an activation date of 2014-02-01 and an unspecified expiration date.

Note: There is no repeating field modifier in front of the opening brace. The equals (=) character is implied, which means the data between the braces replaces the existing Access Level data in the person record.

Example: Delimiters for Two VEHICLES Complex Fields and Subfields

This example shows a repeating list with two **Vehicles** Complex fields separated by the pipe (|) character.



The first **Vehicles** complex field contains data values separated by the tilde (~) character which define:

- **Color:** Silver
- **Make:** Bentley
- **Model:** Arnage
- **State:** MA
- **LicNum:** RICHDUDE
- **TagNum:** Parking permit number, 123

The pipe (|) delimiter indicates that there is an additional **Vehicles** complex field defining another vehicle in the person record.

The second **Vehicles** complex field contains data values for a second vehicle.

Date Fields

Date/Time Field is required in a person record, but if not supplied when adding person records using the **AddPerson** command, the current date is used (date range: 1970 or later).

- **Date** fields are specified in the following format and must be after 1970:
YYYY-MM-DD

Table 10 shows the data fields that require date or date/time formats.

Table 10: Data Fields that Require Date or Date/Time Formats.

Column Keyword	Subfield (if applicable)	Usage Notes
Credentials <i>[list]</i>	ExpirationDate	Date field.
AccessLevels <i>[list]</i>	ActivationDate	Date field. If none specified, or <%BLANK%>, then there is no activation date.
	ExpirationDate	Date field. If none specified, or <%BLANK%>, then no termination date (the person record has an implied "never expires").
ActivationDate ExpirationDate		Date field is required on a person record, but if not supplied on AddPerson, the current date is used (date range: 1970 or later).
LastModDateTime		Date Field (Export and ModifyPerson operation only).

The following example adds the person record of Matthew Yorkin which includes the modification of date and time data in the **Credentials** and **AccessLevels** complex fields.

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME,
CREDENTIALS,ACCESSLEVELS,
AddPerson,Tech Pubs,New Corp,1902,Yorkin,Matthew,{3158~4552~26 bit
Wiegand~Active~2014-02-01~},+{Lab_Pathology~2014-02-01~FALSE~}
```

Using the Exclamation Point (!) for Comments

Column names that start with an exclamation point, '!', such as "!Comment", and their column data are ignored. Otherwise, column names must be keywords.

For example:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,!need to get
FIRSTNAME, CREDENTIALS,ACCESSLEVELS,
AddPerson,Tech Pubs,New Corp,1901,Jones,Need to get first name,
+{3158~4552~26 bit Wiegand~Active~2014-02-01~},+{Lab_Pathology~2014-
02-01~FALSE~}
```

Special Character Handling

The use of quotes is required only for Complex fields.

Table 11 shows how to use quotes in an Import file data row when special characters are included as part of a data field.

Table 11: Special Character Handling.

Special Character	Usage	Example
Pipe ()	If a field is a list and a value has a ' ', put the value in quotes.	"24 7" General Access
Tilde (~)	If a field is complex and a value has a '~', put the value in quotes.	{121~121~"MOT key~fob"~0}
Double-quotes in a quoted value	For any of the above cases, if the string needs to be quoted, data operations checks for quotes in the string to ensure double double-quotes (per Import file).	"General Access" as the value, ""General Access""
Asterisk (*)	An asterisk can be included with the AccessLevels and Credentials complex fields in a data row with a ModifyPerson or ModifyOrAddPerson command. If an access level or credential is defined in your system, but not assigned to the person record being modified, the system will add the access level or credential data. See Using an Asterisk (*) with the AccessLevels and Credentials Complex Fields on page 22 for additional information.	*{Lab_Bio1~2014-10-22 ~~FALSE}

User-Defined Fields

User-defined fields, **UDF1** through **UDF20** are header fields names that correspond to the fields defined on your system (not the actual defined UDF field names on your system).

For example, Figure 103 shows a set of user-defined fields for person records on a system. UDF1 through UDF6 are defined. UDF7 through UDF20 are undefined.

Configure the User-defined Tab

Show?: ☒

Tab Label:

Label*	Enabled	Required	Unique	Editable	Field Size	Field Type	List
Watch This Person	☒	☐	☐	☒	20	Boolean	(none)
DOD Clearance Level	☒	☐	☐	☒	20	List	Clearance
Birthday	☒	☐	☐	☒	20	Numeric	(none)
Years Employed at S2	☒	☐	☐	☒	20	Numeric	(none)
Reports To	☒	☐	☐	☒	20	Text	(none)
Region	☒	☐	☐	☒	20	List	Region
Field7	☒	☐	☐	☒	20	Text	(none)
Field8	☒	☐	☐	☒	20	Text	(none)
Field9	☒	☐	☐	☒	20	Text	(none)

Figure 10: Sample User-Defined fields.

Six user-defined fields are configured:

- **UDF1** = Watch This Person
- **UDF2** = DOD Clearance Level (defined as a list)
- **UDF3** = Birthday
- **UDF4** = Years Employed at LenelS2
- **UDF5** = Reports To
- **UDF6** = Region (defined as a list)
- User-defined fields, DOD Clearance Level and Region, are configured as lists (as shown in Figure).

Access Control | Photo ID | **User-defined*** | Contact | Other Contact | Vehicles | Login | Recent Activity | ?

Watch This Person: ☒

Region:

DOD Clearance Level:

Birthday:

Years Employed at S2:

Reports To:

Field8:

Field9:

Field10:

Field13:

Field14:

Field15:

Field17:

Field18:

Field19:

Field20:

Figure 14: User-Defined fields.

Suppose you need to update a person record because employee George Smith (**Person ID** = 2001) has moved to another region (DC) and is reporting

to a new boss, Al Newboss. He also has updated his security clearance with a new DOD Clearance Level, Top Secret.

To modify his person record, you would use the following header and data rows in your Import file:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,CREDENTIALS,ACCESSLEVELS,UDF1,UDF2,UDF3,UDF4,UDF5,UDF6
ModifyPerson,Tech Pubs,New Corp,2001,{3320~3320~26 bit Wiegand~Active~2014-02-01~~~},{Lab_Pathology~2014-02-01~~FALSE~},TRUE,TopSecret,,,
Al Newboss,DC
```

Partition Fields

Data Operations includes a **Partition** field which allows you to create a person record on a specific partition when operating across multiple partitions. You can also use the **Partition** field to move a person from one partition to another.

For example, the person record of George Smith, who has moved to a new location, is assigned to a new partition, DC Partition.

The header and data rows in the Import file would look like this:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME,CREDENTIALS,ACCESSLEVELS
ModifyPerson,Tech Pubs,DC Partition,2001,Smith,George,
{3320~3320~26 bit Wiegand~Active~2014-02-01~~~},{Lab_Pathology~2014-02-01~~FALSE~}
```

When moving a person from one partition to another, if that person has a user role in the current partition, that role either needs to be removed or the user needs to be assigned a valid user role for the new partition.

Include `<%BLANK%>` in the **LoginUserRole** column of the data row to remove a user role from a person record that is being moved to another partition:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME,
LOGINUSERROLE,CREDENTIALS,ACCESSLEVELS,
ModifyPerson,Tech Pubs,DC Partition,2001,Smith,George,<%BLANK%>,
{3320~3320~26 bit Wiegand~Active~2014-02-01~~~},{Lab_Pathology~2014-02-01~~FALSE~}
```

Access Levels Duplicated Across Partitions

If you attempt to import person record data into a system in which an access level is duplicated across partitions, person records that include an access level with that name will not be imported.

In the Results window, you will see the following error message for each of those person records:

Cannot resolve access level: <accesslevelname>

Global and NetBox Data Operations require different formats for an import operation when an access level is duplicated across partitions:

- Global:

```
{[NetBox name]PartitionName:AccessLevelName~~~}
```

- NetBox:

```
{PartitionName:AccessLevelName~~~}
```

On NetBox, if there are duplicate access levels in the database, the output will be in the form:

```
PartitionName:AccessLevelName.
```

If there are no duplicate access level names in the database, the output will just contain the access level name.

Example: Access Levels Duplicated Across Partitions on an Global System

Suppose you want to import a file on an Global system with an node, MyNetBox, which has two partitions, Partition1 and Partition2. The access level is called Lab_Pathology.

The header row in your Import file would be formatted as follows:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,ACCESSLEVELS,
```

The access level in the data row would include the NetBox name and the partition name:

```
ModifyPerson,MyNetBox,Partition1,2001,+{[MyNetBox] Partition1:
Lab_Pathology~2014-02-01~~FALSE~}
```

Example: Access Levels Duplicated Across Partitions on an NetBox System

Suppose you want to import a file on a NetBox system, MyNetBox, with two partitions, Partition1 and Partition2. The access level is called Lab_Pathology.

The header rows in your Import file would be formatted as follows:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,ACCESSLEVELS
```

The access level in the data row would contain the partition name and would not need to include the NetBox name:

```
ModifyPerson,MyNetBox,Partition1,2001,+{Partition1:Lab_Pathology~201
4-02-01~~FALSE~}
```

Export Files

If you have NetBox systems configured on a NetBox Global system, you can create one Export file for all of them. You will need to create a separate Export file for any NetBox that is not configured on an Global system.

The Data Operations Export feature will not allow you to specify individual person records as you can with the Import feature; it will contain all of the person records. You can, however, choose which of the data fields will be included in the Export file for each person record.

Global and NetBox Export Windows

The Global and NetBox products have slightly different Export windows. The Global system Export window is shown in Figure5. The NetBox system Export window is shown in Figure7.

The following fields appear at the top of the Export window:

- **File Name:** Name of the Export file with the .CSV or .TSV extension.
- **Max Records:** Maximum person records allowed in the file.
- **Partition:** Name of the partition from which the Export file will be retrieved.

Select the partition name from the Partition drop-down menu or select All Partitions.

- **Output Format:** Select CSV or TSV.
- **Open in New Window:** Select this check box to open the contents of the Export file contents in a browser window.
- **Include Images:** Select this check box to include images in the results of an Export operation.

If the Include Images checkbox is selected, the Export operation will produce a zip file containing a comma-separated values (CSV) or tab-separated values (TSV) file.

The file will contain the person record information and a PictureFile field indicating the name of the photo that is assigned to a specific person record.

The name of the zip file will be the name specified in the File Name field of the Export dialog box with its extension changed to zip.

The images are included in the zip file at the top level, so that when the images are extracted from the CSV or TSV file, all of the associated images will be located in the same directory.

Note: The Include Images and Open in New Window options are mutually exclusive.

The screenshot shows the 'Export' dialog box in NetBox Global. The 'File Name' is 'people.csv', 'Output Format' is 'CSV', and 'Open In New Window' is unchecked. 'Partition' is set to 'All Partitions', 'Start Record' is '0', 'Max Records' is empty, and 'Include Images' is unchecked. Under 'Choose headers:', a grid of checkboxes is shown. The following headers are checked: Access Levels, Activation Date, Badge Layout, Credentials, Email, Email2, First Name, Last Name, Login User Name, Login User Role, Other Contact Phone1, Other Contact Phone2, PIN, Phone, UDF1, UDF2, UDF3, UDF4, UDF5, UDF6, UDF8, UDF9, UDF10, UDF11, UDF12, UDF13, UDF14, UDF15, UDF16, UDF17, UDF18, UDF19, UDF20, Use Extended Unlock, and Vehicles. 'Select All' and 'Clear All' buttons are at the bottom left, and 'Ok' and 'Cancel' buttons are at the bottom right.

Figure 15: Global export file header selection.

In Figure 15, the following export header fields have been selected:

- Access Levels
- Activation Date
- Badge Layout
- Credentials
- Email
- Email2
- First Name
- Last Name
- Login User Name
- Login User Role
- Other Contact Phone1
- Other Contact Phone2
- PIN
- Phone
- UDF1 through UDF6

- Vehicles

Figure 116 shows the header and data rows of the Export file displayed in a window on the desktop.

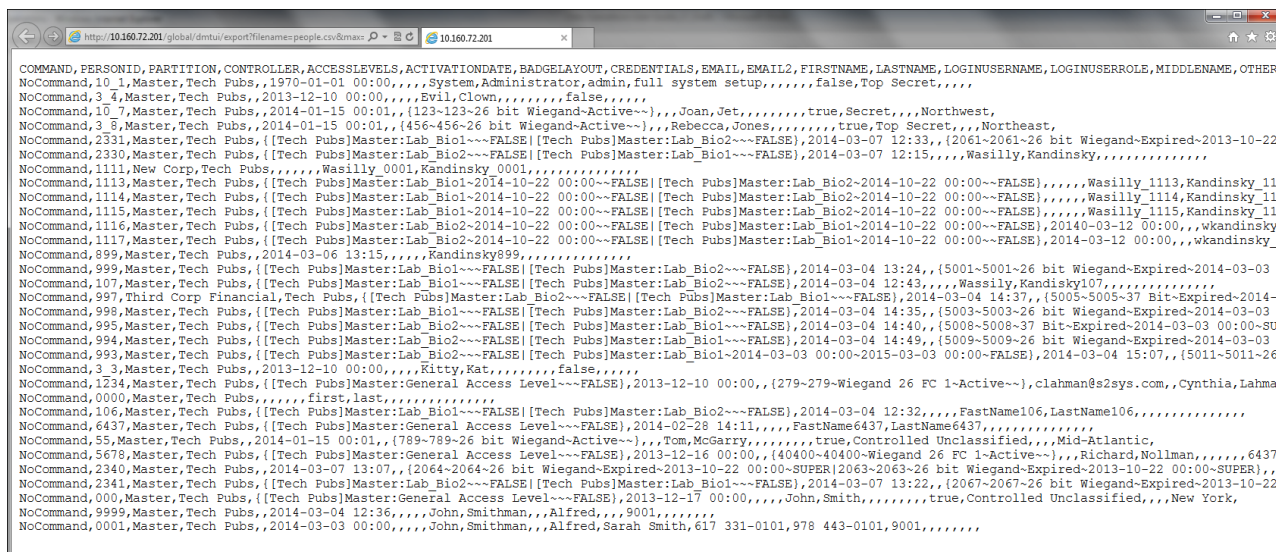
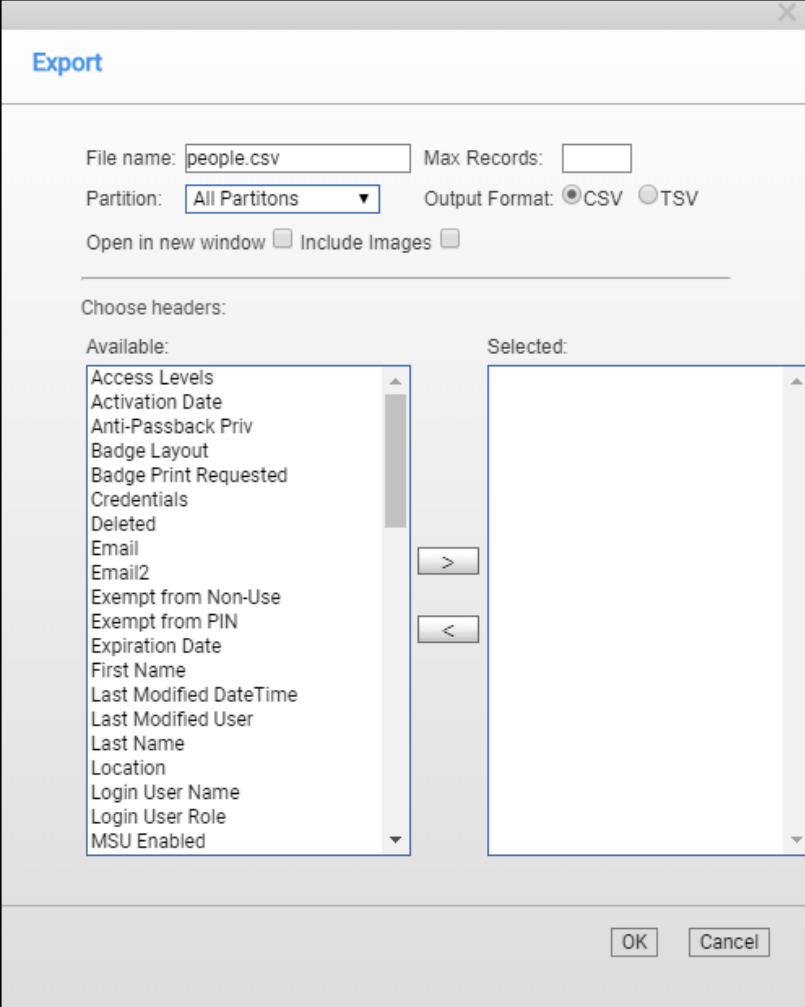


Figure 116: Header and data rows in a sample Export file window.

The Export file is generated as a CSV file by default, or as a TSV file if you selected that format in the Export dialog box.

Note: The NetBox and Global Data Operations user interfaces are identical except for their Export windows.

Figure 7 shows the NetBox system Export window, in which you use the forward (>) and back (<) buttons to select and deselect headers.



The image shows a web-based dialog box titled "Export" with a close button (X) in the top right corner. The dialog contains several input fields and controls:

- File name:** A text input field containing "people.csv".
- Max Records:** An empty text input field.
- Partition:** A dropdown menu currently showing "All Partitons".
- Output Format:** Two radio buttons, "CSV" (which is selected) and "TSV".
- Open in new window:** An unchecked checkbox.
- Include Images:** An unchecked checkbox.
- Choose headers:** A section with two lists and two arrows between them:
 - Available:** A list box containing the following headers: Access Levels, Activation Date, Anti-Passback Priv, Badge Layout, Badge Print Requested, Credentials, Deleted, Email, Email2, Exempt from Non-Use, Exempt from PIN, Expiration Date, First Name, Last Modified DateTime, Last Modified User, Last Name, Location, Login User Name, Login User Role, and MSU Enabled.
 - Selected:** An empty list box.
 - Two arrow buttons, ">" and "<", are positioned between the two lists.
- Buttons:** "OK" and "Cancel" buttons are located at the bottom right of the dialog.

Figure 17: NetBox Export window.

Comparing Data Operations and NetBox Data Management Tool Formats

NetBox Data Management Tool is another tool that allows you to import and export person record data using Import and Export files. In this section, the Import file formats for the NetBox and Global Data Operations feature and the NetBox Data Management Tool are compared.

An Import file for use with the NetBox Data Management Tool contains a header row which specifies the database field names and one or more data rows (one row per person record).

Each data row begins with an API command followed by person record data values. Each field name is separated from the previous one by a tab or comma character.

Field names are not case-sensitive.

Data Management Tool Fields with Multiple Instances

Some Data Management Tool headers can contain multiple instances.

For example, you can define a maximum of 10 credentials for a person record in each row. Credentials are defined by the following headers:

- EncodedNum1 through EncodedNum10
- HotStampNum1 through HotStampNum10
- CardFormat1 through CardFormat10
- CardDisabled1 through Card Disabled10

Figure18 shows an Import file header and data row created using the Data Management Tool, which adds a person record with two credentials for John Adams.

Data Management Headers							
APICommand	PersonID	LastName	FirstName	EncodedNum1	HotStampNum1	CardFormat1	CardDisabled1
AddPerson	1001	Adams	John	111	112	26 bit Wiegand	2020-01-01
Data Management Data				EncodedNum2	HotStampNum2	CardFormat2	CardDisabled2
				222	223	26 bit Wiegand	2020-01-01

Figure 18: Data Management Import file header and data row adding a person record for John Adams with two credentials.

Other header fields that have multiple instances are:

- AccessLevel1 to AccessLevel32
- OtherContPhone1 and OtherContPhone1
- VehicleMake1 and VehicleMake2

- VehicleModel1 and VehicleModel2
- VehicleState1 and VehicleState2
- VehicleLicnum1 and VehicleLicnum2
- VehicleTagnum1 and VehicleTagNum2

See the *NetBox Data Management Tool* guide for additional information on Data Management Import file formats.

Data Operations Complex and Repeating Fields

A Data Operations Import file contains a header row which specifies database field names and multiple data rows (one row per person record). Data Operations also can include complex and repeating fields not supported by the Data Management Tool.

A complex field has multiple subfields separated by a tilde (~) character and is enclosed by opening ({} and closing (}) braces.

The order of subfields is predefined and must appear in the proper order. The **Credentials**, **AccessLevels**, and **Vehicle** headers are complex fields which have specific subfields defined for them. Table 1 on page 12 contains a description of the subfields and their proper order.

A repeating field contains multiple instances in a data row. Repeating fields are separated by the pipe (|) character.

For example, the **Credentials** header has the following header subfields:

- HotStampNum
- EncodedNum
- CardFormat
- Status
- ExpirationDate
- RemoteLockUserType

Figure 19 shows a Data Operations Import file which adds a person record with two credentials for John Adams using complex and repeated fields. Compare this with Figure which shows the similar credentials defined in an Data Management Tool Import file.

Data Management Headers				
Command	PersonID	LastName	FirstName	Credentials
AddPerson	1001	Adams	John	{112~111~26 bit Wiegand~Active~2020-01-01~Regular Access Card 222~223~26 bit Wiegand~Active~2020-01-01~Regular Access Card}
Data Management Data				

Figure 19: Data Operations Import file header and data row adding a person record for John Adams with two credentials.

Configuring Data Operations

You can use your Global and NetBox systems to perform the following Data Operations tasks:

- Upload an Import file.
- Export a file.
- Configure a NAS storage location.
- Schedule an automatic import of a comma-separated values (CSV) or tab-separated values (TSV) file.
- Display detailed information on specific types of errors that were encountered during Import and Export data operations.

The Data Operations user interfaces for NetBox and Global are identical except for some minor differences. These differences are noted in the sections where they apply.

Uploading an Import File

To upload an Import file:

1. In the NetBox web interface, select Administration : Data Operations.
- or -
In the Global web interface, select Setup : Data Operations.
2. Click Upload File.
3. Enter a name for the Import data operation and an optional description.
4. Click **Browse** to open the *File Upload* window.
5. Browse to the file you want to upload.
6. Select the Import file to upload.
7. Click **Open**.
8. (Optional) Select the Prevalidation Max Errors check box to enable Pre-validation of Max Errors and enter a value for Prevalidation Max Errors.

Pre-validation of Max Errors is the number of allowed errors encountered before the Import operation is stopped.
9. Click **OK**.

Data Operations processes the file and displays the results of the Import operation in the Data Operations Results table.

Downloading Results of Past Import Operations

Data Operations provides a list of the current status of jobs. Figure 20 shows the option to download the results of past operations. Use this feature to retrieve errors in an Import file that contains multiple errors. You can use this file to correct errors and then re-submit it as an Import operation.

Click the **Download** button to download the error file (shown in Figure 21).

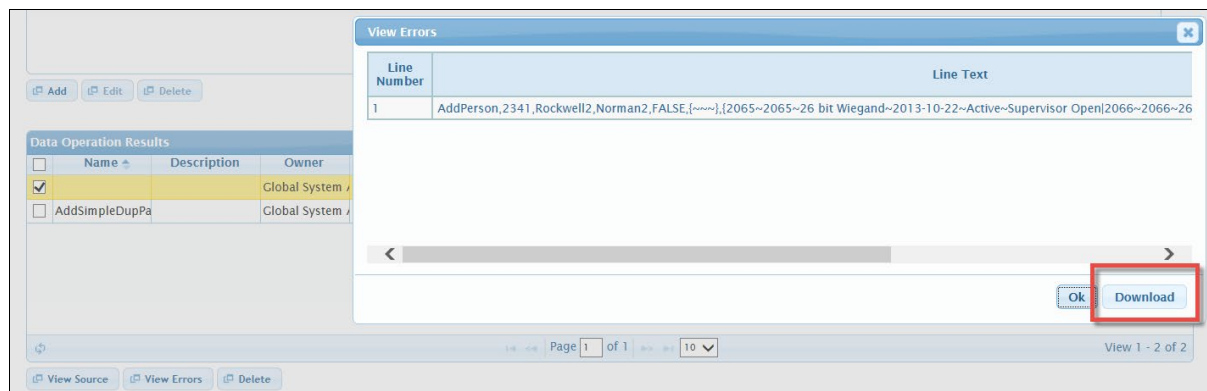


Figure 20: Downloading Data Operations results.

The error file contains the Header and Data rows with the error. You can edit this text file and cut and paste it into an Import file and resubmit it for Data Operations processing.

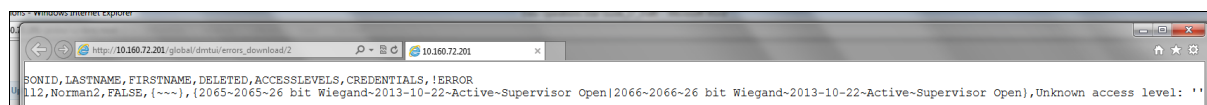


Figure 21: Editing and correcting errors in an error file download.

Configuring a Network Storage Location for Import Operations

To configure network storage:

1. In the NetBox web interface, select Administration: Data Operations.
- or -
In the Global web interface, select Setup: Data Operations.
2. Click **NAS Settings** to open the NAS Settings dialog box.
3. Enter the **Server IP** (address). This is the network IP address of the Server on which the NAS Storage is located.
4. Enter the **Share Name**. This is the name of the network share on the computer on which the NAS Storage is located.
5. Enter a **User Name**, and **Password**: These are required to log into the server.

If the user account is in a Windows domain, use a forward slash to separate the domain name from the username: `<domain>/<username>`.

6. Select the security protocol to be used for authentication when communicating with the remote server. The options are:
 - **None**. Use no authentication (default)
 - **ntlm**. Use NTLM password hashing
 - **ntlmi**. Use NTLM password hashing with signing
 - **ntlmssp**. Use NTLMV2 password hashing encapsulated in Raw NTLMSSP message
 - **ntlmsspi**. Use NTLMV2 password hashing encapsulated in Raw NTLMSSP message and force packet signing
 - **ntlmv2**. Use NTLMv2 password hashing
 - **ntlmv2i**. Use NTLMv2 password hashing with packet signing

Note: In the current release, the selected security protocol is used to protect backups and archives only. It is not used to protect Data Operations import files or report data.

7. Click **OK** to store the NAS configuration or click **Cancel** to abort the operation.

Scheduling an Automatic Import Operation

Using NAS Settings and Automatic Operations mode, a third party can configure the NetBox to poll a network storage (NAS) location for the purpose of performing scheduled automatic Import operations.

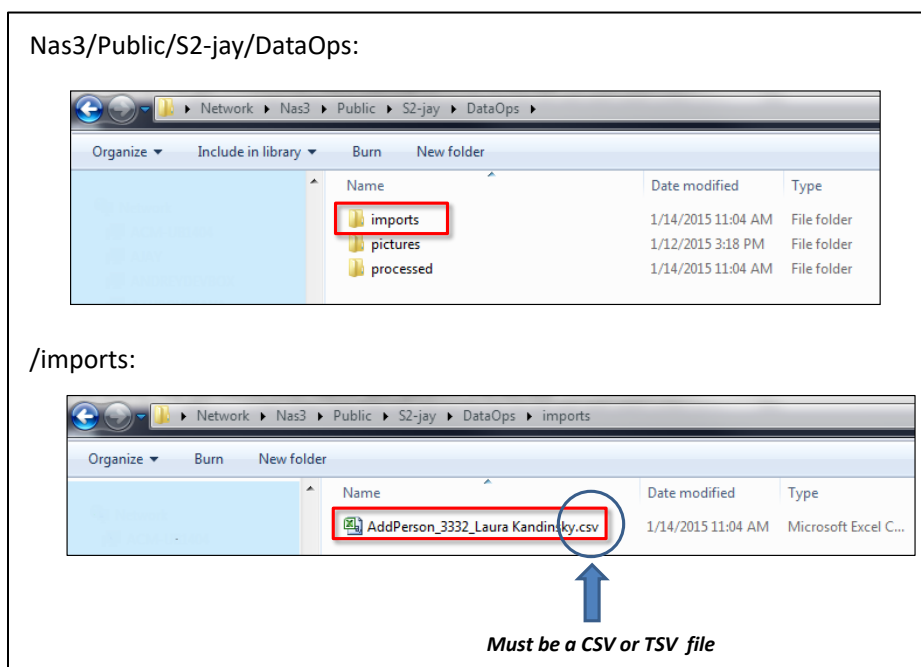
Note: Data Operations can import pictures only through automatic Import operations.

All data import operations work on a NAS location defined on the Network Storage page in the NetBox web interface. The parameters set on this page are used for all data and backup operations.

If a NAS location is not already defined, see the NetBox online help (search for *network storage*) for information on setting up the NAS location.

The NAS base directory must contain the following sub-directories for Data Operations:

- **/imports**: Location where you put files for scheduled import operations.
- **/pictures**: Location where you put image files for scheduled import operations.
- **/processed**: Location where Import files are moved to after being processed by Data Operations.



Note: A picture referenced in the import file must be placed in the */pictures* folder and have either a .JPG or .JPEG extension.

After the import file has been processed, it is placed in the processed directory.

To schedule automatic processing of an Import file from NAS storage:

1. In the NetBox web interface, select Administration : Data Operations.
- or -
In the Global web interface, select Setup : Data Operations.

2. Click **Add**.

For NetBox, the **Add** dialogue box appears. For Global, the **Add Record** dialogue box appears.

3. Enter a name for the automatic Import file operation and an optional description.
4. Enter the NAS directory where the Import file resides.
5. Select an interval from the Interval drop-down menu. Valid values range between 5 minutes and 12 hours for Global and 1 minute and 12 hours for NetBox.
6. (Optional) Select the **Prevalidation Max Errors** check box and enter a value to enable Pre-validation of Max Errors. This is the number of allowed errors encountered before the Import operation is stopped.

7. Click **OK** to engage the operation or **Cancel** to abort it.

Example: Importing Pictures Using an Automatic Import Operation

Suppose you want to add multiple person records that include employee photos to your NetBox. For this example, assume that a NAS location has been set up on a Microsoft Windows computer where the folder C:\S2Backup is being shared:

- Share name: S2Backup
- Directory: S2Backup

To add the person records to your NetBox:

1. Off the NAS share, add three new folders: *Imports*, *Pictures*, and *Processed*. In this example, you will see the following four entries:
 - C:\S2Backup\S2Backup
 - C:\S2Backup\Imports
 - C:\S2Backup\Pictures
 - C:\S2Backup\Processed
2. Create a spreadsheet file for import and save with the suffix .csv.
3. In the spreadsheet file, include rows with the AddPerson command and the data fields for populating the new person records on your NetBox.

Include the PictureFile data field. In that field, add the filename of the picture file for each new person record. You do not need to include the pathname for the picture file.
4. Copy the picture files (.jpg or .jpeg) to the Pictures folder.
5. Copy the spreadsheet file to the Imports folder.
6. In Data Operations, click **NAS Settings** and verify that all fields are correct.
7. In Data Operations, click the **Add** button below the Automatic Operations list.
8. Enter a name for the import operation.
9. Enter a "/" (forward slash character) for the Directory.
10. Choose the interval and select the **Pre-Validation** check box.
11. Click **OK**.

Automatic Operations will display the next run time for the import. When the import operation is complete, an entry will appear in the Data Operations Results list.

12. Verify on your NetBox that all the person records have imported correctly.
13. Delete the automatic import operation from the Automatic Operations list.

Downloading an Export File

To download an Export file from your NetBox system:

1. Select Administration : Data Operations.
2. Click **Export**.
3. Enter an Export file name.
4. Select the output format CSV (comma-separated values) or TSV (tab-separated values).
5. Enter the maximum number of records you want processed in the **Max Records** text box.
6. Select **All Partitions** from the Partition drop-down menu to create an Export file with person records from all the partitions in your NetBox system or select a specific partition.
7. Select the header fields you want to export and click the right-arrow button to move them to Selected list.

The **Command**, **Partition**, and **PersonID** fields will automatically be added to any of the optional fields selected.

(See section: [Import File Format](#) on page 11 for a complete list of data field headers.)

8. Click **OK**.

To download an Export file from your Global system:

1. Select Setup : Data Operations.
2. Click **Export**. The **Export** window opens.
3. Enter a name for the Export file.
4. Fill in the Export window text fields with the appropriate information.
5. Enter the maximum number of records that you want to be processed.
6. Select the **CSV** or **TSV** radio button to specify the type of separators in the file.
7. Select the check boxes of the Header fields for the person record data to be included in the file, or click **Select All** to select for all Header fields.

The **Command**, **Controller**, **Partition**, and **PersonID** fields will automatically be added to any of the optional fields selected.

(See [Import File Format](#) on page 11 for a complete list of data field headers.)

8. Click **OK** to create the Export file.

When the Export operation is completed, you are prompted for the location where the Export file will be saved.

Viewing the Results of Data Operations

The NetBox or Global Data Operations page displays status information on Import data operations and scheduled (automatic) data operations.

Information is displayed on Import data operations in the Data Operations Results table:

- **File Name:** User-defined name of the file.
- **Description:** Textual description of the file.
- **Owner:** User who initiated the operation.
- **Start Time:** Date and time the operation was initiated.
- **End Time:** Date and time the operation was completed.
- **Elapsed Time:** Elapsed time of the operation.
- **Status:** Status of the Import operation:
 - *Complete:* The import operation has completed. Note that completion may occur with errors. The number of errors is shown in the Errors/Lines column.
 - *Validated:* Data in the file has been validated and validation errors have been logged internally.
 - *Processing:* The import operation is still processing.
 - *Waiting:* When multiple import operations have been scheduled, this Import operation is waiting to be processed.
- **Errors/Lines:** Number of recorded error messages in the log file.
- **Failure Info:** Error message text describing the error.
- **Source:** Name of the Import source file.
- **Method:** Upload or automatic.

To delete an entry in the Data Operations Results table:

- In the NetBox web interface, select the **Delete** check box and then click the **Delete** button.
- In the Global web interface, select the row to delete and then click the **Delete** button.

Information on scheduled (automatic) data operations is displayed in the Automatic Operations table:

- **Name:** User-defined name of the file.
- **Description:** Textual description of the data operation.
- **Directory:** Pathname of the Import or Export file.
- **Interval:** Polling interval for the data operation.
- **Pre-Validation:** Whether or not the pre-validation feature is enabled (true=enabled; false=disabled).
- **Max Errors:** Number of recorded error messages in the log file.
- **Owner:** Name of the user who initiated the automatic operation.
- **Last Run Time:** Time the last scheduled automatic operation was run.
- **Next Run Time:** Time that the next scheduled automatic operation was run.
- **Status:** Status of the automatic operation is either "*Submitted*" or an error message is displayed. See [Error Messages](#) on page 51 for details.

Error Messages

Global and NetBox data operations log the following error types:

- File Processing: For example, *No COMMAND* in header
- Syntax: For example, *Missing* trailing quote
- Row Processing: For example, *Wrong number of columns*

Viewing Global Import File Errors

To view Import file errors on your Global system:

1. In the Data Operations Results window, select the check box for a specific Import operation in the list of currently available data operations.

The selected operation row is highlighted in yellow:

Data Operation Results											
☐	Name	Description	Owner	Start Time	End Time	Elapsed Time	Status	Errors/Lines	Failure Info	Source	Method
☒			Global System	03/26/2014 17:31:0	03/26/2014 17:31:0	0:00	Complete	1 / 1		AddPerson_2340_S	upload
☐	AddSimpleDupPa		Global System	03/21/2014 15:21:4	03/21/2014 15:21:4	0:00	Complete	1 / 1		AddPerson_Simple	upload

Page 1 of 1 | 10 | View 1 - 2 of 2

View Source View Errors Delete

2. Select the View Errors button to display the error message shown below:

Add Edit Delete

☐	Name	Description	Owner
☒			Global System
☐	AddSimpleDupPa		Global System

View Errors

Line Text	Error
i bit Wiegand~2013-10-22~Active~Supervisor Open 2066~2066~26 bit Wiegand~2013-10-22~Active~Supervisor Open	Unknown access level: "

Ok Download

Page 1 of 1 | 10 | View 1 - 2 of 2

View Source View Errors Delete

Error Processing and Log Files

When errors are encountered in an Import file, a separate CSV file is constructed which includes all rows that had an error. A column is appended to the CSV file, called: "!Error", which contains the error message.

When Import file data rows that do not contain errors are processed, the corresponding person records are added, deleted, or updated in your system. Data rows with errors are reported in the !Error file.

For example, the Import file shown in Figure 1222 contains errors in the second and last data rows.

The error in the second data row contains a typo. "Lab_Bio30," which should be Lab_Bio3." In our example, the data row is processed and Lab_Bio30 access level is not found on the system.

The last data row has a duplicate person ID. After successfully adding a person record for Dave Ford with person ID 2017, the system cannot add another person record for Alfred Steele with the same person ID.

The errors messages for these two data rows are appended to the !Error file.

The other data rows do not contain errors, so they are processed and the person records are added to the system as shown in Figure 1222.

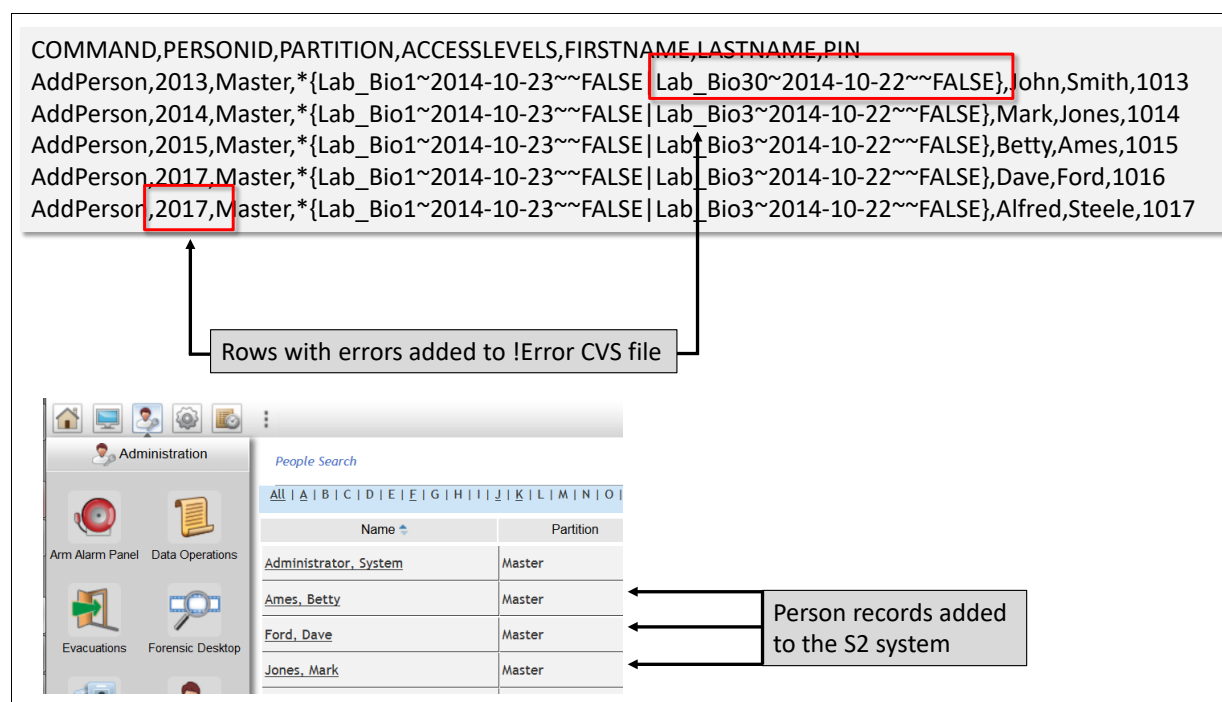


Figure 12: Data Operations error processing.

Examples of Import Files with Errors

Here are some examples of Import files with errors:

```
COMMAND,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME, NOTES,!Error
ModifyPerson,Tech Pubs,New Corp,12345,<%BLANK%>,<%BLANK%>,My Notes,Person
record must at least have one of last name or first name
```

```
ADDPERSON,CONTROLLER,PARTITION,PERSONID,LASTNAME,FIRSTNAME,
ACCESSLEVELS,!Error
AddPerson,Tech Pubs,New Corp,12346,Doe,John,{LAX ACCESS},Access level does not
exist on target
```

Data Operations Error Messages

Table 12 shows the Line and File errors reported by NetBox and Global data operations.

Table 12: Data Operations Error Messages.

Error Message	Comments
/mnt/remotes/dmtmnt/'Directory'/pictures/<filename> (No such file or directory)	This error will occur for one of the following reasons: - Data Operations could not find the image file in the <i>/pictures</i> sub-directory. - Data Operations could not find the <i>/pictures</i> sub-directory. See Scheduling an Automatic Import Operation on page 45 for information on how to set up NAS directories for Data Operations.
A deleted person may not be assigned credentials.	You cannot update a credential for a person record that has been deleted.
Anti Passback Priv not found: <value>.	Value is not one of the pre-defined values for the AntiPassBackPriv field: Exempt, Soft Always, and Hard Always. For example: If the value present in the Import file for the AntiPassBackPriv field is Never, Data Operations would present the following error message: <i>Anti Passback Priv not found: Never</i>
Assignment already exists for <value>.	A complex field with a "+" modifier already exists.
Badge layout file not found: <value>.	Value specified in the BadgeLayout field is not known for the person's partition.
Bad header.	The header row is not defined properly. Check for: - Incorrect spelling of keywords - Missing or extraneous delimiters
Cannot resolve access level: <value>.	The Access Level specified in the data row was not found. Check to see if the access level exists on the system you have specified and if the spelling is correct.

Error Message	Comments
Color value <value> greater than 20 characters.	Value specified in the Vehicles (header field), Color (subfield) is greater than the maximum (20) characters allowed.
Controller and/or partition need to be specified.	There was not enough information to uniquely identify where to add the person in the AddPerson command. A partition is sufficient if it is unique; otherwise you also need to specify the controller name. Supplying the controller only is sufficient if you wish to default to the master partition.
Duplicate credential. Hot stamp #: <value> , Encoded #: <value> .	The credential supplied already exists in the system.
Duplicate person ID: <value> .	The person ID supplied already exists in the system.
Duplicate user name: <value> .	The user name supplied already exists in the system. Login names must be unique.
Email value <value> greater than 50 characters.	Value specified in the Email header field is greater than the maximum (50) characters allowed.
Encoded number <value> is too large for selected format.	Enter a valid encoded number.
Error mounting NAS directory.	Incorrect IP address, username, password, or share name.
Error obtaining file lock.	This error may occur when there are permission or contention issues with the Import file directory. When multiple import operations have been scheduled, this Import operation may be waiting to be processed.
Execution stopped after <count> errors.	Processing stopped after number of errors in the error count. If this error count is encountered after the pre-validation phase, all prior data is committed to the system. If the error count is reached during pre-validation, no data has been written.
FIPS card format requires 14 digit encoded numbers.	Enter a valid FIPS card format.
First or last name must not be null or empty.	A person record must include either FirstName or LastName .
Imports directory does not exist.	Data Operations cannot find the Imports directory. See Scheduling an Automatic Import Operation on page 45 for information on how to set up NAS directories for Data Operations.
Invalid activation date. Must be after 01-Jan-1964.	A valid activation date must be greater than 01-Jan-1964.
Invalid column name: <value> .	Column names must be valid keyword headers. Enter an exclamation point (!) leading character for this field name to be explicitly ignored.

Error Message	Comments
Invalid command: <value> .	Command keyword must be one of the following: • AddPerson • ModifyPerson • ModifyOrAddPerson
Invalid complex field format for <value> .	The syntax for the complex field specified is not formatted correctly. See Import File Format on page 11 for information on Import file syntax.
Invalid controller and partition.	The Controller , Partition or combination do not yield a valid partition target (Global Only).
Invalid controller name: <value> .	The Controller name does not exist. (Global Only).
Invalid credential profile: <value> .	Enter a valid credential profile.
Invalid date format for <field> .	See Date and Time Fields on page 32 for details on formatting.
Invalid encoded number: <value> .	Enter a valid encoded number.
Invalid expiration date. Must be after 01-Jan-1964.	A valid expiration date must be greater than January 1, 1964, specified as 1964-01-01.
Invalid file format.	Enter a valid file format.
Invalid partition name: <value> .	The partition specified in the Import file is not configured on your system.
Invalid remote lock set user type: <value> .	The remote lock set user type is not defined on your system.
Invalid role for user: <value> .	The User role specified in the Import file is not configured for the specific person record on your system.
Invalid role name: <value> .	The specified role name is not configured on your system.
Invalid state code: <value> .	A standard two-character state code (such as MA) was not specified in the State keyword header sub-field in a Vehicles complex field.
Invalid true/false value for <field> .	A Boolean value ("true" or "false") is required.
Last name must be specified.	A last name was not specified in the LastName column.
License# value <value> greater than 20 characters.	Value specified in the Vehicles (header field), Licnum (subfield) is greater than the maximum (20) characters allowed.
Location value <value> greater than 30 characters.	Value specified in the Location (header field) is greater than the maximum (30) characters allowed.
Make value <value> greater than 20 characters.	Value specified in the Vehicles (header field), Make (subfield) is greater than the maximum (20) characters allowed.
Model value <value> greater than 20 characters.	Value specified in the Vehicles (header field), Model (subfield) is greater than the maximum (20) characters allowed.
Notify on Expirations Group not found: <value> .	A Notify on Expirations email distribution group not defined for the person record.

Error Message	Comments
No match found on delete for <value>.	The attempted deletion failed because no matching value was found.
Partition name <value> is not unique, controller field must be specified.	The controller must be specified for fields that require a partition qualifier, or for the Partition field, if partition name is not unique.
Password must not be null or empty.	You must supply a password in this field.
Person ID must be specified.	The PersonID field is required. It is the unique identifier for the person you are adding or modifying. A PersonID header and data field must be supplied in an AddPerson and ModifyPerson command.
PersonID not found: <value>.	No person record with the person record ID number was found.
Phone value <value> greater than 20 characters.	Value specified in the Phone (header field) is greater than the maximum (20) characters allowed.
Phone 2 value <value> greater than 20 characters.	Value specified in the OtherContactPhone1 or OtherContactPhone2 header field is greater than the maximum (20) characters allowed.
Required column <value> missing from header.	Enter required column value.
STRAC 128 bit card format requires 32 hex (0-F) characters.	Enter the correct card format.
Syntax: Closing quote is not the last character. Syntax: Missing trailing quote. Syntax: No <character> after closing quote. Syntax: Unexpected quote.	The row or field being processed expects a closing quote. See Special Character Handling on page 33 for details on how Data Operations handles quotes.
Tag# value <value> greater than 20 characters.	Value specified in the Vehicles (header field), Tagnum (subfield) is greater than the maximum (20) characters allowed.
The <value> card format is not enabled.	The card format needs to be enabled in the system user interface. To enable the format: • Navigate to the Card/Keypad Formats page (<i>Configuration : Access Control : Card/Keypad Formats</i>). • Select the card format from the drop-down menu. • Select Enabled checkbox for the card format. • Click Save.
The <value> card status is not enabled.	Enable the card format.
The value <value> is not a number.	Enter a number.
UDF<udfnum1-N> is not enabled.	This UDF field must be enabled as follows: • Navigate to the Card/Keypad Formats page (<i>Configuration : Access Control : Person Records : Configuration</i>). • Select the Enabled checkbox for the UDF field.

Error Message	Comments
	Select Save.
UDF<udfnum1-N> is required.	This UDF field is required by the system and must be specified. The AddPerson command must define it. The ModifyPerson command cannot set it to NULL.
UDF<udfnum1-N> list value does not include <value>.	This UDF field is of type "list" and the value specified is not a valid value for the list as defined in the system.
UDF<udfnum1-N> value must be a number.	This UDF field is of type "numeric" and non-numeric values were provided.
UDF<udfnum1-N> value must be 'true' or 'false': <value>.	This UDF field is Boolean ("true" or "false") and a non-Boolean value was specified.
UDF<udfnum1-N> value must be unique.	This UDF field requires that each person record has a unique value.
Unknown access level: <value>.	The AccessLevelName value in the AccessLevels field is not in the system.
Unknown anti-passback privilege: <value>.	A valid value was not found. Valid values for NetBox are: Exempt, Soft Always, and Hard Always. Valid values for Global are: Exempt, Soft, and Hard.
Unknown card format: <value>.	The CardFormat name value in the Credentials field is not in the system.
Unknown card status: <value>.	The Status name value in the Credentials field is not in the system.
User role must be specified with user name and password.	Specify user role with user name and password.
Wrong number of columns.	The number of columns for this row does not match the number of header keyword fields. See Import File Format on page 11 for information on Import file syntax.
Wrong number of fields for <complex field>. Expected <count>, actual <count>.	The number of fields for this complex field is in error. See Import File Format on page 11 for information on Import file syntax.

Appendix A - Attribute Maximum Field Length (Person/Vehicle)

As specified by the NetBox DataOps, the Person and Vehicle attribute fields come with a maximum character limit to ensure smooth mapping and synchronization processes. Please do not surpass the specified character count to maintain seamless functionality.

Person Attribute Field	Maximum Characters
Phone	20
MobilePhone	20
Email	50
Email2	50
Location	30

Vehicle Attribute Field	Maximum Characters
Color	20
Make	20
Model	20
Licnum	20
Tagnum	20